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Two-stage non-cooperative games with risk-averse players

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Abstract This paper formally introduces and studies a non-cooperative multi-agent game under uncertainty. The well-known Nash equilibrium is employed as the solution concept of the game. While there are several formulations of a stochastic Nash equilibrium problem, we focus mainly on a two-stage setting of the game wherein each agent is risk-averse and solves a rival-parameterized stochastic program with quadratic recourse. In such a game, each agent takes deterministic actions in the first stage and recourse decisions in the second stage after the uncertainty is realized. Each agent's overall objective consists of a deterministic first-stage component plus a second-stage mean-risk component defined by a coherent risk measure describing the agent's risk aversion. We direct our analysis towards a broad class of quantile-based risk measures and linear-quadratic recourse functions. For this class of non-cooperative games under uncertainty, the agents' objective functions can be shown to be convex in their

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own decision variables, provided that the deterministic component of these functions have the same convexity property. Nevertheless, due to the non-differentiability of the recourse functions, the agents' objective functions are at best directionally differentiable. Such non-differentiability creates multiple challenges for the analysis and solution of the game, two principal ones being: (1) a stochastic multi-valued variational inequality is needed to characterize a Nash equilibrium, provided that the players' optimization problems are convex; (2) one needs to be careful in the design of algorithms that require differentiability of the objectives. Moreover, the resulting (multi-valued) variational formulation cannot be expected to be of the monotone type in general. The main contributions of this paper are as follows: (a) Prior to addressing the main problem of the paper, we summarize several approaches that have existed in the literature to deal with uncertainty in a non-cooperative game. (b) We introduce a unified formulation of the two-stage SNEP with risk-averse players and convex quadratic recourse functions and highlight the technical challenges in dealing with this game. (c) To handle the lack of smoothness, we propose smoothing schemes and regularization that lead to differentiable approximations. (d) To deal with non-monotonicity, we impose a generalized diagonal dominance condition on the players' smoothed objective functions that facilitates the application and ensures the convergence of an iterative best-response scheme. (e) To handle the expectation operator, we rely on known methods in stochastic programming that include sampling and approximation. (f) We provide convergence results for various versions of the best-response scheme, particularly for the case of private recourse functions. Overall, this paper lays the foundation for future research into the class of SNEPs that provides a constructive paradigm for modeling and solving competitive decision making problems with risk-averse players facing uncertainty; this paradigm is very much at an infancy stage of research and requires extensive treatment in order to meet its broad applications in many engineering and economics domains.

Mathematics Subject Classification 91A10 Non-cooperative games · 91A15 Stochastic games · 91A20 Multi-stage games

1 Introduction

Since George B. Dantzig's formulation in the mid 1950s [13] of linear programming under uncertainty, the field of stochastic programming has blossomed into a fruitful subject in optimization, supported by significant theoretical and algorithmic advances over the last half-century with numerous texts, such as [4,28,33,34,63,71], and scholarly articles as well as dedicated activities within professional societies; see the Committee on Stochastic Programming website http://stoprog.org/. As is well known, stochastic programming (SP) deals with a class of optimization models and algorithms in which some of the data may be subject to significant uncertainty. Such models are appropriate when data evolve over time, and decisions need to be made prior to observing the entire data stream. For instance, investment decisions in portfolio planning must be implemented before stock performance can be observed. Similarly, utilities must plan power generation and/or capital investments before the demand for

electricity is realized. Such inherent uncertainty is amplified by technological innovation and market forces. Despite the vast amount of research on this subject, the SP framework quickly becomes inadequate when multiple players are present in the system, who for various reasons make decisions independently of each other. This could be due to the lack of a centralized coordination mechanism among these players, or to insufficient communication among them possibly because of their spatial separation, or to the selfish behavior of each that dictates their non-cooperation and leads to the sub-optimality of the overall system welfare. For instance, power producers in electricity markets make bids contingent on their existing fuel-price contracts and cost structure of their generation assets in the face of uncertain demand and generation availability, the latter a consequence of the growing proportion of renewable energy sources. Unfortunately, extant SP models and algorithms are not designed to accommodate multiple decision makers who do not cooperate as a unity in such a non-cooperative multi-agent system.

Studied largely under deterministic assumptions, non-cooperative game theory also has a long history; this theory deals with the resolution of conflicts among multiple selfish players each of whom has an optimization problem to solve that is parameterized by the rivals' decision variables. Unlike the SP framework, however, deterministic game theory fails to capture the adaptive decision-making structure that is core to the handling of uncertainty. Providing a solution concept to a game, a Nash equilibrium (NE) [47,48] is by definition a tuple of strategies, one for each player, such that no player will be better off by unilaterally deviating from his/her equilibrium strategy while the rivals keep executing their equilibrium strategies. Although it was first noted by Bensoussan [2] that the problem of finding a NE can be formulated as a variational inequality, the contemporary treatment of the NE, through finite-dimensional variational inequalities (VIs) and complementarity problems (CPs) gained momentum after the publication of the survey [26] which was later expanded into the 2-volume monograph [16]. One of the more important consequences of the above development is its manifestation in effective computational schemes for NE. Two recent surveys on Nash equilibria via the variational approach are [15,17] where many references can be found. In contrast to the more than 50 years of research of stochastic programming that deals with single-agent optimization problems under uncertainty and the advances on the Nash equilibrium problem (NEP) for competitive multi-agent optimization in the last ten years or so, the study of the NEP under uncertainty is, at best, at a stage of infancy. Today, with the two subjects of stochastic programming and deterministic NEP at a fairly mature state, the time is ripe to undertake a comprehensive investigation of stochastic Nash equilibrium problems (SNEPs). As one can expect from the deterministic NEP, there are many issues arising from the SNEP that distinguish it from a SP and make it a worthwhile effort for an in-depth investigation.

There are several common types of uncertainties that may arise in a SNEP. One type of SNEP involves making decisions (such as capital investments) at the present time to meet the uncertainty (such as demands) that are revealed at a later time; with the first-stage decisions in place, the second-stage decisions are the activity levels (such as production) to meet the realized uncertainty. This is the well-known paradigm of two-stage stochastic programming with recourse, generalized to a competitive system where multiple decision makers are planning their strategies in the face of their

rivals' counter actions. A related type of uncertainty arises in a two-settlement market consisting of a deterministic (first-stage) spot market and a stochastic (second-stage) market known as the forward; in this situation financial contracts are written at the present time to guard against adverse realizations of future events. Another type of uncertainty arises when the economic agents are risk-averse and they aim to minimize their risks as they make their operation/investment decisions (see [36]). Each agent's risk aversion is quantified by a *coherent risk measure* as formalized by [59,60]. The two-stage Nash game with risk-averse players, which is the main problem to be studied in this paper, provides a broad mathematical framework for treating these problems.

As expected, applications of the SNEP have begun to emerge in recent years. These include [21] that discusses a scenario-based dynamic oligopolistic problem under uncertainty, [14,23–25] that discuss capacity expansion in electricity markets, [36] that addresses supply-side risks, [35,77] that discuss two-settlement markets with the uncertainty in the future market, and [20] that presents a stochastic complementarity model of equilibrium in an electric power market, in the presence of uncertainty about power demand. Most recently, the paper [65] presents Nash equilibrium models of perfectly competitive capacity expansion involving risk-averse participants in the presence of discrete state uncertainties and pricing mechanisms of different kinds. In addition to the modeling and the derivation of economic insights, several of the cited papers [7,20,35,36,65] also address properties of the models, such as existence of an equilibrium, and propose computational methods for solving the resulting models; see also [37,78] that discuss the resolution of smooth and nonsmooth stochastic Nash games via sampling and smoothing. To date, much of the analysis and solution methods for solving the risk-averse SNEP is based on the single-valued variational inequality formulation of the game, which as we will explain subsequently is not fully justified in the presence of non-differentiable recourse functions in the players' objective functions. Most recently, extending the previous work [44], the paper [45] considers equilibrium problems for a market with competing risk-averse agents subject to coupling constraints resulting from regulatory interventions or market clearing conditions. The latter paper employs smoothing of the risk functions but does not discuss in detail the numerical solution of the resulting smoothed games; it relies on the solver PATH that is principally for solving deterministic VI/CPs. In contrast, our work here goes far deeper into the combination of deterministic methods with sample approximations. Incidentally, the paper [46] discusses a smoothing sample average approximation method for stochastic optimization problems with the CVAR risk measure.

King and Rockafellar [39] first considered generalizations of the variational inequality problem to the regime where the components of the map are expectation-valued. Existence statements [54,55] and sampling-based schemes [37,71,78] for such problems have seen much study (see [70] for a review). More recently, the subject of (single-valued) stochastic variational inequalities (SVI) has been given a fundamental treatment through the consideration of multi-stage generalizations [61]; compared to previous approaches, the derivation of the SVI framework in the cited reference makes intuitive sense and serves as the foundation for future research on the problem. The cornerstone of the framework is the very important non-anticipativity concept that has played a central role in stochastic programming to date. In order for the risk-averse SNEP to fit into the SVI framework as set forth in [61], a first requirement is that

the players' objective functions need to be differentiable; this is not automatic without some smoothing and regularization as will be discussed in Sect. 6. Even within the SVI framework, there remains the important issue of the numerical solution of the resulting formulation; this is not addressed in [61] and much work is needed to make this framework a practically viable approach. In this paper, we bypass the SVI framework and deal with the risk-averse SNEP based largely on the best-response scheme whose effectiveness has been amply demonstrated in the deterministic setting, for example to multiuser telecommunication systems [17,67]. Supplementing this work, we develop in [68] a distinctly different approach for addressing a class of *single-stage* stochastic Nash games with expectation-valued objectives. This reference considers an inexact best-response scheme (that utilizes stochastic approximation techniques) and presents rate statements and an overall iteration-complexity analysis.

In general, best-response algorithms are quite natural for solving non-cooperative games in that they allow the individual players to independently update their strategies in response to their rivals' strategies. Mathematically, the convergence of this family of algorithms requires conditions that have an intuitive appeal; roughly speaking, these conditions stipulate that the impact of the individual players' own decisions on their respective objectives dominates the collective impact of their rivals' strategies on the objectives. Most importantly, due to the generality of our framework, no existing algorithm with supporting convergence theory and completely implementable steps is directly applicable to the game with its detailed ingredients that include risk measures of a general nature, two-stage decision making, and quadratic recourse functions. Our goal in this paper is to offer a foundation to the formulation of the resulting two-stage game with such ingredients and its solution via sampling and best response. Since there are many more details involved in the practical implementation of the algorithms to be presented, we will leave their application to the domain problems mentioned before to subsequent research. Our expectation of the numerical performance of the algorithms is that they will provide a promising venue for solving two-stage games with risk-averse players taking first- and second-stage decisions; nevertheless, further research is needed to bring the benefits and impact of this work to its full fruition.

2 Stochastic game-theoretic models

There are several formulations of a non-cooperative game under uncertainty; we briefly present three of them—the risk-neutral game, the robust game, and an almost-sure game—before we introduce the main model to be treated in this paper, namely the two-stage game with risk-averse agents. The purpose of presenting the alternative models of the SNEP is not so much as to attempt to advance their theory or computation, which is clearly impossible given the length limit of the article, but rather to give a review of a few existing approaches to deal with non-cooperative games under uncertainty.

2.1 The risk-neutral game

By way of introduction, consider a non-cooperative game with N risk-neutral players, each of whom, labeled $i = 1, \ldots, N$, has a private strategy set $X^i \subseteq \mathbb{R}^{n_i}$ that is closed and convex, a (deterministic) first-stage objective function θ_i that depends on

all players' strategies $x \triangleq \left(x^j\right)_{j=1}^N \in \mathbf{X} \triangleq \prod_{j=1}^N X^j$, and a second-stage expected recourse function $\varphi_i(x) = \mathbb{E}[\psi_i(x, \widetilde{\omega})]$, where the random recourse function $\psi_i(x, \widetilde{\omega})$ not only depends on x but also on the realization of a random variable $\widetilde{\omega}$ defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with Ω being the sample space, $\mathcal{F}$ being the σ-algebra generated by subsets of Ω, and $\mathbb{P}$ being a probability measure defined on $\mathcal{F}$. [Throughout the paper, the tilde on $\widetilde{\omega}$ signifies a random variable, whereas ω without the tilde refers to a realization of the variable.] This setting includes a finite scenario model where the sample space Ω is finite. Anticipating the rivals' strategy $x^{-i} \in X^{-i} \triangleq \prod_{i \neq j=1}^N X^j$, player i chooses an optimal response by solving the following stochastic program with parameter x^{-i}:

$$\underset{x^i \in X^i}{\text{minimize}}\ \zeta_i(x^i, x^{-i}) \triangleq \theta_i(x^i, x^{-i}) + \underbrace{\mathbb{E}\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right]}_{\varphi_i(x^i, x^{-i})}. \tag{1}$$

A Nash equilibrium (NE) is a tuple $x^* \triangleq \left(x^{*,j}\right)_{j=1}^N \in \mathbf{X}$ such that $x^{*,i} \in \underset{x^i \in X^i}{\text{argmin}}\ \zeta_i(x^i, x^{*,-i})$ for all $i = 1, \ldots, N$. As in SP, the recourse function $\psi_i(x; \omega)$ is the (optimal) value function of a second-stage optimization problem parameterized by the pair $(x; \omega)$,

$$\psi_i(x; \omega) \triangleq \underset{z^i \in Z^i(x;\omega)}{\text{minimum}}\ f_i(z^i; x; \omega) \tag{2}$$

where $Z^i(x; \omega)$ is a closed convex subset of $\mathbb{R}^{m_i}$ and $f_i(\bullet; x; \omega) : \mathbb{R}^{m_i} \to \mathbb{R}$ is a given scalar-valued function. For large classes of optimization problems (e.g. linear programs), value functions such as $\psi_i(\bullet; \omega)$ may not be differentiable as it is well known in sensitivity theory of linear and nonlinear programs. Indeed, due to the likely presence of inequality constraints in the set $Z^i(x; \omega)$ involving the x-variable, the value function $\psi_i(x; \omega)$ is at best piecewise smooth, and thus not differentiable, in x (see [6] for an extensive study on perturbation analysis of optimization problems and [16, Chapter 5] for a broad sensitivity analysis of variational inequalities under perturbation); this piecewise property carries over to the expected value function $\mathbb{E}\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right]$. Indeed, as proved in the pioneering work of Wets [75], this expected value is continuous when the distribution function of the random variable $\widetilde{\omega}$ is continuous and the recourse function (2) is a linear program; more generally, the absolute continuity of the random variable $\widetilde{\omega}$ is needed as a sufficient condition for such a differentiability property [32]. In this sense, approaches that rely on the differentiability of the recourse functions may fail to provide any guarantee for models of this genre. In the event where $\psi_i(\bullet; \omega)$ is differentiable, the formula: $\nabla_{x^i} \mathbb{E}\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right] = \mathbb{E}\left[\nabla_{x^i} \psi_i(x^i, x^{-i}; \widetilde{\omega})\right]$ is guaranteed under the assumptions of the classical Lebesgue dominated convergence theorem that ensures the interchange of integration and differentiation. In the more general case with $\psi_i(\bullet, x^{-i}; \omega)$ being convex, the formula $\partial_{x^i} \mathbb{E}\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right] = \mathbb{E}\left[\partial_{x^i} \psi_i(x^i, x^{-i}; \widetilde{\omega})\right]$ requires a technical measurability condition; see [62,71].

The above definition of the game where the players have only private constraints given by the sets X^i and objectives given by $\zeta_i(x)$ with coupled variables adheres to the standard formulation of a non-cooperative game. In defining the recourse function $\psi_i(x;\omega)$, however, we have included the rivals' variables in both the objective function $f_i(\bullet; x; \omega)$ and feasible set $Z^i(x;\omega)$. In order to ensure the finiteness of the recourse, this generality implicitly imposes some restrictions on the pair $(x;\omega)$ in the form of *coupled constraints*, making the overall game one of the generalized kind [17,52,66]. Indeed, the optimization problem (1) is equivalent to:

$$\underset{x^i \in \widehat{X}^i(x^{-i})}{\text{minimize}}\ \zeta_i(x^i, x^{-i}) \triangleq \theta_i(x^i, x^{-i}) + \mathbb{E}\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right] \tag{3}$$

where

$$\widehat{X}^i(x^{-i}) \triangleq \left\{ x^i \in X^i \mid -\infty < \mathbb{E}\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right] < \infty \right\}, \tag{4}$$

is the rival-dependent constraint set for player i; the expected-value constraints in $\widehat{X}^i(x^{-i})$ couple all the players' variables, thus the term "coupled constraints". This is an important aspect of the SNEP that is not present in a deterministic game; namely, in the latter game, if there are only private constraints on the agents' variables, no coupled constraints will ever be introduced. Nevertheless, in the SNEP, coupled constraints on the agents' variables could arise from potential infeasibility or unboundedness implicit in the second-stage problem, which in turn would result in implied constraints (such as feasibility cuts) coupling variables of all players. Needless to say, if the recourse function $\psi_i(x;\omega)$ is such that $\widehat{X}^i(x^{-i}) = X^i$ for all i, then the constraint coupling disappears and the "generalized" game becomes a standard game. This is the case of focus in our later treatment.

2.2 The robust game

This provides a worst-case approach to model uncertainty and is based on the concept of a robust solution to an optimization problem with an *uncertainty set* [3]. While some references [1,27,49,50] have addressed the resulting games, the treatment therein is rather restrictive, relatively speaking. Moreover, these references follow the traditional form of robust optimization to avoid having an expectation (since this requires access to a probability measure) and instead employ a worst case formulation through an uncertainty set. To broaden the treatment, we adopt the minmax stochastic programming formulation for the players' optimization problems. Thus, our formulation is a little different in that we employ both an expectation and an uncertainty set. Specifically, consider an N-person game where the individual players' (parametric) optimization problems are:

$$\left\{ \underset{x^i \in X^i}{\text{minimize}}\ \zeta_i(x^i, x^{-i}) \triangleq \left\{ \theta_i(x^i, x^{-i}) + \underset{u^i \in \mathcal{U}^i}{\text{maximum}}\ \mathbb{E}\left[g_i(x^i, x^{-i}; u^i; \widetilde{\omega})\right] \right\} \right\}_{i=1}^{N} \tag{5}$$

with the set $\mathcal{U}^i$ and the function $g_i(x; u^i; \widetilde{\omega})$ being player dependent (as indexed by i). This formulation is sufficiently broad to include the case of a discrete maximum where $\mathcal{U}^i$ is the unit simplex, or more generally, the convex hull of finitely many vectors. Mathematically, due to the generality of the formulation, the above abstract minmax framework (5) includes the reformulated two-stage SNEP derived from a broad class of risk/deviation measures after a suitable algebraic manipulation of the latter; see the game $\mathcal{G}^{\text{rfMD}}$ in Sect. 4.4. Nevertheless, unlike the two-stage SNEP with recourse to be introduced in Sect. 4, only a highly conservative treatment of risk through the worst case maximization is accounted for in (5); moreover, the notion of recourse is at best hidden in the function g_i.

The game defined by (5) differs from the type discussed in [1] in that this reference is concerned primarily with finite-action and finite-scenario games, resulting in the players' expected payoffs being multilinear functions of their decision variables that are discrete probabilities of the mixed strategies. This is a much more restrictive class of games than the above formulation where the players' strategy sets X^i can in principle be any closed convex sets. The games discussed in [27,49,50] focus on special types of objectives (e.g. quadratic) and uncertainty sets (e.g. defined by Euclidean and Frobenius norms); for these specialized robust games, the authors were able to reformulate them as cone-complementarity problems, thereby allowing their solution by semidefinite complementarity methods.

As in the case of the recourse function (2), the overall objective function ζ_i in (5) is not differentiable even if the functions θ_i and g_i are; as such, a direct treatment of this N-person game by VI theory that requires differentiability is again not possible. As a way to deal with the non-differentiability induced by the max operator, we may adopt a *pull-out* approach proposed in [18] for treating games with minmax objective functions. Specifically, by considering a $2N$-person game where player $i = 1, \ldots, N$ solves the optimization problem with primary variable x^i and parameters (x^{-i}, u^i),

$$\underset{x^i \in X^i}{\text{minimize}}\ \theta_i(x^i, x^{-i}) + \mathbb{E}\left[g_i(x^i, x^{-i}; u^i; \widetilde{\omega})\right],$$

and player $N + j$ for $j = 1, \ldots, N$ solves the optimization problem in the variable u^j with parameter $x \in \mathbf{X}$,

$$\underset{u^j \in \mathcal{U}^j}{\text{maximize}}\ \mathbb{E}\left[g_j(x; u^j; \widetilde{\omega})\right].$$

Note that each of these $2N$ problems is a stochastic program. Based on [18, Remark 1], it is not hard to show that a NE of the latter $2N$-player game is a NE of the original robust N-person game. In the deterministic case, or even in the finite-scenario case, this pull-out approach has the advantage of allowing the sets $\mathcal{U}^i$ to be fairly general, as long as they render the resulting $2N$-person game amenable to a constructive treatment; moreover, the non-differentiability issue of the original N-person game is resolved provided that the functions θ_i and g_i are all differentiable and that the expectation can be interchanged with differentiation. Such an interchange is possible if e.g., either the sample space is finite or some conditions on the random functions $g_j(x; u^j; \bullet)$ are in place for the applicability of the Lebesgue dominated convergence theorem.

It should be noted that as pointed out in [18], to date the numerical solution of the pulled-out game is not easy except in the case where the functions g_i and uncertainty sets $\mathcal{U}^i$ are the same across all the players. In general, smoothing via convolution could be applied to the original N-person game with minmax stochastic player problems (see for instance [78]). The detailed solution of this game is nevertheless beyond the scope of this paper.

2.3 An almost-sure model

Another approach to model uncertainty leads to what we may call an *almost-sure Nash Equilibrium Problem* (asNEP). While retaining the stochastic nature of the equilibrium, the concept of recourse is absent in this almost-sure formulation which seeks "good" approximations to almost every scenario-realization of the problem. Specifically, the asNEP seeks a (deterministic) tuple x^* such that for almost every $\omega \in \Omega$, x^* is a Nash equilibrium solution of the N-person game wherein player i's optimization problem is

$$\underset{x^i \in X^i}{\text{minimize}}\ \widehat{\theta}_i(x^i, x^{-i}; \omega), \tag{6}$$

where $\widehat{\theta}_i$ is a given player-dependent random function. Assuming that $\widehat{\theta}_i(\bullet, x^{-i}; \omega)$ is convex and its gradient exists $\nabla_{x^i}\widehat{\theta}_i(\bullet, x^{-i}; \omega)$ exists (for simplicity), letting $\mathcal{N}(x; \mathbf{X})$ denote the normal cone of the set $\mathbf{X}$ at the vector $x \in \mathbf{X}$, and using the generalized equation formulation of the first-order optimality conditions of the above optimization problems (6) for all $i = 1, \ldots, N$, we see that x^* is an almost-sure Nash equilibrium if and only if $\mathbb{P}\{\omega \in \Omega \mid 0 \in \Theta(x^*; \omega) + \mathcal{N}(x^*; \mathbf{X})\} = 1$, where $\Theta(x^*; \omega) \triangleq \left(\nabla_{x^i}\widehat{\theta}_i(x^*; \omega)\right)_{i=1}^N$. Unless in a pathological setting, it is highly unlikely that a deterministic x^* will satisfy the generalized equation $0 \in \Theta(x^*; \omega) + \mathcal{N}(x^*; \mathbf{X})$ for almost every ω. This motivates the use of a residual formulation of the inclusion $0 \in \Theta(x; \omega) + \mathcal{N}(x; \mathbf{X})$, for fixed ω in terms of a nonsmooth equation, such as that based on the natural map of the corresponding VI; see [16] for details. In general, suppose $H(x; \omega) = 0$ is an equation formulation for the VI [16, Volume II]. [An example would be $H(x; \omega) \triangleq x - \Pi_{\mathbf{X}}(x - \Theta(x; \omega))$, where $\Pi_{\mathbf{X}}$ is the Euclidean projector onto the set $\mathbf{X}$.] Consider the expected residual minimization problem:

$$\underset{x \in \mathbf{X}}{\text{minimize}}\ \zeta(x) \triangleq \mathbb{E}\left[\, \| H(x; \widetilde{\omega}) \| \,\right], \tag{7}$$

where $\| \bullet \|$ is an arbitrary vector norm. A global minimizer of (7) with a zero objective value is an (exact) almost-sure Nash equilibrium. Since ζ is in general a nonconvex function, it is unrealistic to expect that such a global minimizer can be obtained in practice. Thus, focus is placed on the analysis and computation of a stationary solution of (7) which may be considered a surrogate almost-sure Nash equilibrium. In essence, this is the approach of expected residual minimization adopted in the work of [8,9,19,29,42] that pertains to the so-called almost-sure VI. The drawback of this approach is that there is no feedback relation of the computed solution to an equilibrium solution

of the SNEP. Furthermore, the example of the function H given above makes it clear that even the computation of a stationary solution of (7) is not easy and requires much further research in order for this approach to be practically appealing.

In summary, the above discussion has introduced 3 formulations of a stochastic Nash equilibrium problem under uncertainty. Starting in the next section, we introduce the main model of the SNEP that will be the focus of the rest of the paper.

3 The risk-averse model

The two-stage SNEP with risk-averse players is formally defined as follows. Player i employs a deviation measure $\mathcal{D}_i(\mathcal{Z})$, where $\mathcal{Z} : \Omega \to \mathbb{R}$ denotes a random variable defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Note the player dependency of such a measure, reflecting the distinctive risk aversion of the individual player/firm. The ith player solves the following parameterized risk-averse minimization problem

$$\underset{x^i \in X^i}{\text{minimize}}\ \zeta_i(x^i, x^{-i}) \triangleq \theta_i(x^i, x^{-i}) + \underbrace{\mathbb{E}\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right] + \lambda_i\, \mathcal{D}_i\left[\psi_i(x^i, x^{-i}; \widetilde{\omega})\right]}_{\text{denoted } \mathcal{R}_{i,\lambda_i}(x^i, x^{-i})}, \tag{8}$$

where $\lambda_i \in [0, \infty)$ is a nonnegative weight reflecting a player-dependent trade-off between the expected value and the deviation measure. We call $\mathcal{R}_{i,\lambda_i} : \mathbf{X} \to \mathbb{R} \cup \{\pm\infty\}$ a *mean-deviation (or risk) composite function*, where the emphasis is on the composition $(\mathbb{E} + \lambda_i \mathcal{D}_i) \circ \psi_i(x; \bullet)$ of the risk function $\mathcal{R}_{i,\lambda_i} \triangleq \mathbb{E} + \lambda_i \mathcal{D}_i$ with the (recourse) function $\psi_i(x; \omega)$. From a modeling perspective, the two-stage game with risk aversion is quite different from an uncertainty-set based robust game. For one thing, the concept of an optimization-based recourse function is typically not accounted for in the formulation of the latter game, and moreover, the worst-case scenario expressed by the pointwise maximum over all elements of the uncertainty set (cf. the max term in (5)) is only one kind of risk aversion as compared to the risk composite function $\mathcal{R}_{i,\lambda_i}$ which accommodates other risk measures. Mathematically, one could perhaps relate the two formulations—risk aversion versus robust—more closely; but such a relation is clearly beyond the scope of this paper.

The connections between deviations and risks have been well explained in the references [56,59,60]; but their compositions with a recourse function seem to have attracted less attention. By definition, a deviation measure is a nonnegative convex operator. Some well-known deviation measures include the standard deviation, lower and upper semideviations, mean absolute (semi)deviations, absolute (semi)deviations, the median, and deviations derived from the conditional value-at-risk (CVAR); see [59,60]. All these measures are intrinsically non-differentiable, resulting in the optimization problem (8) being non-smooth in principle. This non-smoothness complicates the numerical solution of the game, especially if one desires to design provably convergent distributed algorithms so that each player updates his/her strategy independently of the rivals.

It should be remarked that even though a deviation measure is a convex operator as implied by the axiomatic properties of the measure, such properties alone are in general not enough to ensure the convexity of the composite function $\mathcal{R}_i(x) \triangleq \mathcal{D}_i \circ \psi_i(x; \widetilde{\omega})$ in x, which is the primary decision variable of the players of the game. Since $\psi_i(x; \omega)$ is typically at best piecewise smooth, and thus nonlinear, in x, care is needed in adopting the deviation measure in order to ensure that convexity (at least in the players' individual variables) is preserved in the resulting mean-deviation composite function $\mathcal{R}_{i;\lambda_i}(x^i, x^{-i})$. An additional monotonic property on $\mathcal{D}_i$ suffices for this purpose. The paper [64] has discussed extensively the optimization of convex risk functions that is closely related to the optimization of mean-deviation objectives such as (8). In Sect. 4, we will focus on a large class of mean-quantile deviations that admit a minmax expected-value, and thus a convex formulation; see the Ph.D. thesis [74] and Chapter 6 of the lecture notes [71] for such optimization problems. A more general discussion of the convexity of risk composite functions and their optimization can be found in the latter set of lecture notes and in the earlier paper [64]. In particular, [71] provides a rich resource of the contemporary foundation of optimization under risk.

As already mentioned, the game with the player problems given by (8) is potentially of the generalized type when the mean-deviation composite function $\mathcal{R}_{i,\lambda_i}(x^i, x^{-i})$ is extended-valued. In the subsequent treatment of this problem, we will impose conditions on the recourse function and the deviation measure to ensure the finiteness of $\mathcal{R}_{i,\lambda_i}(x)$ for all $x \in \mathbf{X}$, thus removing the generalized nature of the game.

4 Mean-deviation-composite games with quadratic recourse

Historically, the development of stochastic programming began with the case where the recourse function is defined by a linear program (LP) parameterized by the pair $(x; \omega)$. Even today, this paradigm of LP-based recourse functions still occupies a substantial literature of SP research. As a generalization of this classic paradigm and to prepare for a smoothing of the LP-based recourse, we begin our study of the risk-averse SNEP by considering a convex quadratic program (QP) based recourse function. Our path to the treatment of the resulting class of games is as follows. After presenting a QP-based recourse function and summarizing its key properties, we introduce the class of quantile-based deviation measures that leads to a reformulation of the mean-deviation optimization problems (8) in terms of expectations. This reformulated game, which we denote by $\mathcal{G}^{\text{rfMD}}$ provides the basis for the development of computational algorithms for solving the original game. Non-smoothness and other major challenges remain in $\mathcal{G}^{\text{rfMD}}$ in general; Sect. 5 introduces a special case of this game where the recourse functions are all *private*; i.e., each does not contain the rival players' variables. We show that this special game, which we denote by $\mathcal{G}^{\text{prMD}}$, can be readily solved by an iterative best-response algorithm requiring the solution of standard mean-deviation stochastic programs with quadratic recourse. Assuming exact solutions of the latter SPs, the convergence of the algorithm can be established without too much difficulty based on approaches in the existing game-theoretic literature; see [17]. Nevertheless, this algorithm is useful only in the situation when the sample space Ω is finite and the probabilities of the discrete scenarios are known so that the expectation operator can

be written as a finite sum. More generally for the game $\mathcal{G}^{\text{rfMD}}$ with coupled constraints in the recourse functions, we can apply smoothing and regularization procedures to induce differentiability of the mean-deviation composite functions. Convergence of the equilibria of the smoothed games to an equilibrium of the original game is established using the theory of uniform convergence, which provides a convenient way to verify the advanced multi-epi-convergence theory [25]. These procedures are discussed in detail in Sect. 6 which also describes sample-average approximation based best-response algorithms for solving the smoothed games.

4.1 QP-based recourse functions

Consider the players' optimization problems as given by (8) with the recourse function $\psi_i(x;\omega)$ being the optimal value of a convex quadratic program parameterized by the pair $(x;\omega)$:

$$\begin{aligned} \psi_i(x;\omega) \triangleq \underset{z^i}{\text{minimum}} & \left[f^i(\omega) + \underbrace{\sum_{j\neq i} G^{ij}(\omega)x^j}_{\text{no } x^i} \right]^T z^i + \tfrac{1}{2}(z^i)^T Q^i z^i \\ \text{subject to } & \underbrace{\sum_{j=1}^{N} C^{ij}(\omega)x^j}_{\text{contains } x^i} + D^i z^i \geq \xi^i(\omega), \end{aligned} \tag{9}$$

where Q^i and D^i are constant matrices with Q^i being symmetric positive semidefinite, f^i and ξ^i are two random vector functions, and each G^{ij} (for $i \neq j$) and C^{ij} are random matrix functions, all defined on the sample space Ω and of appropriate orders. This setting implicitly involves a fixed recourse assumption in that D^i is a deterministic matrix for $i = 1, \ldots, N$. Notice that x^i does not appear in the objective function in (9). Otherwise, the convexity of $\psi_i(x^i, x^{-i};\omega)$ in x^i will be in jeopardy. Quadratic recourse functions have been studied in the SP literature; references include: [43,69] that discussed cutting-plane methods for SPs having such recourse functions with [43] discussing mainly the case of discrete random events; [10,11,40] that presented test problems and discussed non-smooth Newton methods for solving discrete approximations of such SPs formulated as convex programs; [5] whose focus was to formulate a model to induce certain smoothness of the recourse function; and [38] that developed an L-shaped algorithm for solving SPs with quadratic recourse in a limited setting. Besides the game-theoretic aspect, a major departure of our work from these references is that the quadratic recourse functions are combined with a risk function and best-response algorithms are introduced to solve the resulting games.

Under a *relatively complete recourse* assumption, i.e., if $\psi_i(x;\omega)$ is finite for all pairs $(x;\omega) \in \mathbf{X} \times \Omega$ and $\mathbb{E}[\psi_i(x;\widetilde{\omega})]$ is bounded, then provided that the deviation measure $\mathcal{D}_i(\mathcal{Z})$ is finite valued for all random variables $\mathcal{Z}$ with finite expectations,

we have $\widehat{X}^i(x^{-i}) = X^i$ for all $x^{-i} \in X^{-i}$. As an initial undertaking of a detailed study of the two-stage stochastic game with risk-averse players, the relatively complete recourse assumption seems reasonable as a way to simplify the analysis; future research can relax this assumption and deal with other generalizations.

While this issue is classical in the literature of SP, see e.g. [73], we present a self-contained result (Proposition 2) that summarizes key properties of the mean-deviation composite function $\mathcal{R}_{i,\lambda_i}(x)$; in turn, this result relies on some basic properties of the optimal solutions to a convex quadratic program which we collect in Lemma 1. This lemma is reminiscent of the pioneering work of Wets [72,75,76] on solution properties of linear and (convex) quadratic programs; see also [22] for a related early work. A contemporary comprehensive study of the qualitative properties of solutions to (convex) quadratic programs that contains more general results is provided by the monograph [41]. A couple of noteworthy remarks about the lemma below are: (i) the matrix Q is assumed to be positive semidefinite but not positive definite, and (2) the statements are global in nature and pertain to all solvable QPs with fixed matrices but variable data vectors in the objective and constraint. For completeness and self-containment, we give a detailed proof of each part of the lemma.

Lemma 1 *Let $Q \in \mathbb{R}^{n\times n}$ be a symmetric positive semidefinite matrix and $D \in \mathbb{R}^{m\times n}$ be an arbitrary matrix. For each vector $\xi \in \mathbb{R}^m$, let $S(\xi) \triangleq \{z \in \mathbb{R}^n \mid Dz \geq \xi\}$ denote the feasible region of the following convex quadratic program:*

$$\underset{z\in S(\xi)}{\text{minimize}}\ f^T z + \tfrac{1}{2} z^T Q z. \tag{10}$$

Further, let $S^{\text{opt}}(f;\xi)$ and $qp_{\text{opt}}(f;\xi)$ denote the solution set and the optimal value of (10), respectively. Then, the following statements hold:

(a) *the set Ξ of vector pairs $(f;\xi) \in \mathbb{R}^{n+m}$ for which (10) has a finite optimal objective value is a polyhedron; moreover, the vector Qz and the scalar $f^T z$ are constants, denoted respectively by $QS^{\text{opt}}(f;\xi)$ and $f^T S^{\text{opt}}(f;\xi)$, on the optimal solution set $S^{\text{opt}}(f;\xi)$; the scalar $z^T Qz$ is also constant on this set;*
(b) *the mapping $(f;\xi) \in \Xi \mapsto QS^{\text{opt}}(f;\xi)$ is a piecewise linear, thus Lipschitz continuous function; moreover, for every fixed $f \in \mathbb{R}^n$, so is the mapping $\xi \mapsto f^T S^{\text{opt}}(f;\xi)$ provided that $(f,\xi) \in \Xi$;*
(c) *the constant $z^T Qz$ is a continuous function of the pair $(f,\xi) \in \Xi$, where $z \in S^{\text{opt}}(f,\xi)$; thus so are the mappings $(f;\xi) \in \Xi \mapsto \text{qp}_{\text{opt}}(f,\xi)$ and $(f;\xi) \in \Xi \mapsto f^T S^{\text{opt}}(f;\xi)$;*
(d) *a constant $\eta > 0$ exists such that for all $(f;\xi) \in \Xi$,*

$$|\,\text{qp}_{\text{opt}}(f;\xi)\,| \leq \eta \left[\|\xi_+\| \,(\|f\| + \|Q\|\,\|\xi_+\|) + \|Q\|\,\|f\|^2 \right];$$

[in particular, if $0 \in S(\xi)$, then $|\text{qp}_{\text{opt}}(f;\xi)| \leq \eta\,\|Q\|\,\|f\|^2$]
(e) *the mapping $(f;\xi) \in \Xi \mapsto qp_{\text{opt}}(f;\xi)$ is Lipschitz continuous on every compact subset $\widehat{\Xi} \subseteq \Xi$.*

Proof **(a)** The set Ξ consists of pairs $(f;\xi)$ for which there exist $z \in S(\xi)$ and $\lambda \geq 0$ satisfying $f + Qz - D^T\lambda = 0$. This establishes the polyhedrality of Ξ. The singleton property of $QS^{\text{opt}}(f;\xi)$ and $f^T S^{\text{opt}}(f;\xi)$ is well known in convex quadratic programming. The constancy of $z^T Qz$ for $z \in S^{\text{opt}}(f;\xi)$ follows from that of Qz for $z \in S^{\text{opt}}(f;\xi)$ and can be proved as follows. Write $Q = RR^T$. [This decomposition is valid for a symmetric positive semidefinite Q. If Q is singular, so is R; the singularity does not affect the proof below.] We claim that $R^T S^{\text{opt}}(f;\xi)$ is a singleton. Indeed, let z and z' be any two vectors in $S^{\text{opt}}(f;\xi)$. We have $RR^T(z-z') = 0$, which yields $(z-z')^T RR^T(z-z') = 0$. Thus $R^T z = R^T z'$. This establishes of the constancy of $z^T Qz$ for $z \in S^{\text{opt}}(f;\xi)$.
(b) This follows from [16, Exercises 5.6.14 and 5.6.15].
(c) To prove this part, we write the necessary and sufficient conditions of the QP (10) as follows:

$$\begin{array}{l} 0 = f + Qz - D^T v \\ 0 \leq -\xi + Dz \;\perp\; v \geq 0, \end{array} \tag{11}$$

which yield a couple of consequences:

1. the multi-valued KKT map: $S^{\text{KKT}} : \Xi \mapsto S^{\text{KKT}}(f;\xi) \triangleq \{(z,v) \mid (z,v) \text{ satisfies (11)}\}$ is a polyhedral multifunction; i.e., its graph is the union of finitely many polyhedra; as such, $S^{\text{KKT}}(f;\xi)$ is pointwise upper Lipschitz continuous in (f,ξ); i.e., for every (f,ξ), there exists a constant $L > 0$ and a neighborhood $\mathcal{N}$ of (f,ξ) such that for every $(f';\xi') \in \mathcal{N}$,

$$S^{\text{KKT}}(f';\xi') \subseteq S^{\text{KKT}}(f;\xi) + L\left[\,\|f'-f\| + \|\xi'-\xi\|\,\right]\mathbb{B},$$

   where $\mathbb{B}$ is the unit ball in the (z,v)-space;
2. there exists a constant $c^{Q,D} > 0$, dependent on the pair (Q,D) only, such that for every $(f,\xi) \in \Xi$, a pair (z,v) satisfying (11) exists such that

$$\|(z;v)\| \leq c^{Q,D}\left[\,\|f\| + \|\xi\|\,\right]. \tag{12}$$

[Remark: Due to the positive semidefiniteness of Q, the sets $S^{\text{KKT}}(f;\xi)$ and $S^{\text{opt}}(f;\xi)$ are actually polyhedra; nevertheless, using the respective polyhedral representations of these sets of KKT-pairs and optimal solutions will yield a constant c dependent on the pair $(f,\xi) \in \Xi$; such a dependence is precisely what we want to avoid. This will become clear in the following proof of continuity of the optimal objective value; see also Proposition 3.] To prove the continuity of the optimal objective value $\text{qp}_{\text{opt}}(f;\xi)$, let $\{(f^k,\xi^k)\} \subset \Xi$ be a sequence of vector pairs converging to $(f^\infty;\xi^\infty)$. Since Ξ is closed, it follows that $(f^\infty;\xi^\infty) \in \Xi$. For each k, let $(z^k;v^k)$ be a pair satisfying the KKT system (11) corresponding to (f^k,ξ^k) such that $\|(z^k;v^k)\| \leq c^{Q,D}\left[\|f^k\| + \|\xi^k\|\right]$, and let $(\bar{z}^k;\bar{v}^k) \in S^{\text{KKT}}(f^\infty;\xi^\infty)$ be such that

$$\|(z^k;v^k) - (\bar{z}^k;\bar{v}^k)\| \leq L\left[\,\|f^k - f^\infty\| + \|\xi^k - \xi^\infty\|\,\right].$$

Since the sequence $\{(z^k; v^k)\}$ is bounded, so is the sequence $\{(\overline{z}^k; \overline{v}^k)\}$; moreover, the limit points of these two sequences are the same. Let $(z^\infty; v^\infty)$ be the limit of a convergent subsequent $\{(z^k; v^k)\}_{k\in\kappa}$ of the former sequence for an infinite subset κ of the iterate counters; then $(z^\infty; v^\infty)$ is also the limit of the subsequence $\{(\overline{z}^k; \overline{v}^k)\}_{k\in\kappa}$. Hence $(z^\infty; v^\infty) \in S^{\mathrm{KKT}}(f^\infty; \xi^\infty)$ and $z^\infty \in S^{\mathrm{opt}}(f^\infty; \xi^\infty)$. Moreover,

$$\begin{aligned}\lim_{k(\in\kappa)\to\infty} \mathrm{qp}_{\mathrm{opt}}(f^k; \xi^k) &= \lim_{k(\in\kappa)\to\infty} \left[(f^k)^T z^k + \tfrac{1}{2} (z^k)^T Q z^k \right] \\ &= (f^\infty)^T z^\infty + \tfrac{1}{2} (z^\infty)^T Q z^\infty \;=\; \mathrm{qp}_{\mathrm{opt}}(f^\infty, \xi^\infty)\end{aligned}$$

establishing the convergence of the sequence of objective values $\{\mathrm{qp}_{\mathrm{opt}}(f^k; \xi^k)\}$ to $\mathrm{qp}_{\mathrm{opt}}(f^\infty; \xi^\infty)$. The same proof also establishes the continuity of the constant $z^T Q z$ as a function of (f, ξ) for $z \in S^{\mathrm{opt}}(f; \xi)$. This establishes part (c).

(d) In essence, this part can be proved by applying (12); however, we are interested in obtaining a sharper bound that involves $\xi_+ \triangleq \max(\xi, 0)$, which is the nonnegative part of ξ; thus the following alternative derivation. By the renowned Hoffman error bound for polyhedra, it follows that there exists a constant $\alpha > 0$, depending on D only, such that for any ξ for which $S(\xi) \neq \emptyset$, a vector $z^{\mathrm{ref}} \in S(\xi)$ exists such that $\|z^{\mathrm{ref}}\| \leq \alpha \|\xi_+\|$. Hence,

$$\mathrm{qp}_{\mathrm{opt}}(f; \xi) \;\leq\; f^T z^{\mathrm{ref}} + \tfrac{1}{2} (z^{\mathrm{ref}})^T Q z^{\mathrm{ref}} \;\leq\; \alpha \,\|f\|\, \|\xi_+\| + \frac{\alpha^2}{2} \|Q\|\, \|\xi_+\|^2 \tag{13}$$

To complete the proof, consider the Lagrangian dual of the QP (10) (see [12, p. 117]):

$$\begin{aligned}&\underset{z,v}{\text{maximize}}\; \xi^T v - \tfrac{1}{2} z^T Q z \\ &\text{subject to } -Qz + D^T v \;=\; f \text{ and } v \;\geq\; 0.\end{aligned} \tag{14}$$

Similar to the above, there exists a constant $\beta > 0$ such that for every vector f for which the above dual is feasible, a feasible pair $(z^{\mathrm{feas}}, v^{\mathrm{feas}})$ exists such that $\|(z^{\mathrm{feas}}, v^{\mathrm{feas}})\| \leq \beta \|f\|$. Since v^{feas} is nonnegative, we have

$$\begin{aligned}\mathrm{qp}_{\mathrm{opt}}(f; \xi) &\geq \xi^T v^{\mathrm{feas}} - \tfrac{1}{2} (z^{\mathrm{feas}})^T Q z^{\mathrm{feas}} \\ &\geq -(\xi_+)^T v^{\mathrm{feas}} - \frac{\beta^2}{2} \|Q\|\, \|f\|^2 \\ &\geq\; -\beta \|f\| \left[\|\xi_+\| + \frac{\beta}{2} \|Q\|\, \|f\| \right].\end{aligned}$$

Combining this inequality with (13), we easily derive the existence of a constant $\eta > 0$ so that (d) holds.

(e) The proof of this part follows from the results in [41, Section 16.1] where the objective value map $\mathrm{qp}_{\mathrm{opt}}$ is shown to be a continuous piecewise linear-quadratic function on Ξ (the continuity is also proved in part (c) above); i.e., there exist finitely

many quadratic function $\{q_i(f;\xi)\}_{i=1}^K$ for some positive integer $K > 0$, such that $\mathrm{qp}_{\mathrm{opt}}(f;\xi) \in \{q_i(f;\xi)\}_{i=1}^K$ for all $(f;\xi) \in \Xi$. As such this map is locally Lipschitz on Ξ [16, Lemma 4.6.1]. [A mapping Θ is *locally Lipschitz* on a domain $\mathcal{D}$ if for every $x \in \mathcal{D}$, a neighborhood $\mathcal{O}_x \subseteq \mathcal{D}$ of x exists such that Θ is Lipschitz continuous on $\mathcal{O}_x$.] Therefore, it remains to show that every locally Lipschitz continuous function on a compact subset must be Lipschitz continuous on the set. For this purpose, let Θ be locally Lipschitz continuous on the compact set S. Assume by way of contradiction that Θ is not Lipschitz continuous on S. Thus, for every positive integer k, there exists a pair of vectors (x^k, y^k) in S such that

$$|\,\Theta(x^k) - \Theta(y^k)\,| \;>\; k\,\|\,x^k - y^k\,\|,\ \forall\, k \;=\; 1, 2, \ldots. \tag{15}$$

The compactness of S leads to two observations: (1) the absolute value on the left-hand side is bounded; thus the sequence $\{x^k - y^k\}$ converges to zero as $k \to \infty$; (2) the sequence $\{x^k\}$ has a convergent subsequence $\{x^k\}_{k\in\kappa}$ with limit x^∞. Thus $\{y^k\}_{k\in\kappa}$ also converges to x^∞. By the locally Lipschitz property of Θ near x^∞, a constant $c_\infty > 0$ exists such that for all $k \in \kappa$ sufficiently large, $|\,\Theta(x^k) - \Theta(y^k)\,| \le c_\infty \|\,x^k - y^k\,\|$. But this contradicts (15). □

It is useful to remark that Lemma 1 makes no assumption about the boundedness of either the feasible set $S(\xi)$ or the optimal set $S^{\mathrm{opt}}(f;\xi)$. Another noteworthy remark is that the Lemma does not assert the joint Lipschitz continuity of the optimal objective function on the full set Ξ if it happens to be unbounded.

Extending the well-known case of LP-based recourse (see e.g. [71, Chapter 2]), Proposition 2 below makes a relatively complete recourse assumption under which the finiteness and a boundedness property of the optimal objective value $\psi_i(x;\omega)$ given by (9) can be established. Under this assumption, the graph of the set-valued map $\widehat{X}^i : X^{-i} \to X^i$ is equal to the set $\mathbf{X}$ thus $\widehat{X}^i(x^{-i}) = X^i$ and we are back to the basic case with no coupled constraints. For a random variable $\mathcal{Z}$, we recall that

$$\mathrm{essup}(\mathcal{Z}) \;\triangleq\; \inf\{\,\alpha \;\mid\; \mathbb{P}\{\mathcal{Z} \;\mid\; \mathcal{Z} > \alpha\} = 0\,\}$$

is the *essential supremum* of $\mathcal{Z}$. We say that a random variable $\mathcal{Z}$ is *essentially bounded* if $\mathrm{essup}(|\mathcal{Z}|) < \infty$.

Proposition 2 *Let Q^i be a symmetric positive semidefinite matrix. Assume that*

- *the (random) matrix functions $G^{ij}(\widetilde{\omega})$ and $C^{ij}(\widetilde{\omega})$ and the (random) vector functions $f^i(\widetilde{\omega})$ and $\xi^i(\widetilde{\omega})$ are all measurable and bounded on Ω such that*

$$\max\left(\left\{essup\ |G^{ij}(\widetilde{\omega})|\right\}_{i\neq j},\ \left\{essup\ |C^{ij}(\widetilde{\omega})|\right\}_{i,j=1}^N,\right.$$
$$\left.\left\{essup\ |f^i(\widetilde{\omega})|,\ essup\ |\xi^i(\widetilde{\omega})|\right\}_{i=1}^N\right) \;<\; \infty; \tag{16}$$

- *for almost all $\omega \in \Omega$ and all $x \in \mathbf{X}$,*

$$\left[(D^i)^T v^i = 0,\ v^i \geq 0 \right] \Rightarrow \left[\sum_{j=1}^{N} C^{ij}(\omega) x^j - \xi^i(\omega) \right]^T v^i \geq 0,$$

$$D^i z^i \geq 0,\ Q z^i = 0 \Rightarrow \left[f^i(\omega) + \sum_{j \neq i} G^{ij}(\omega) x^j \right]^T z^i \geq 0; \quad (17)$$

- *the following bound holds:*

$$essup \left\{ \sup_{x \in \mathbf{X}} \left[\xi^i(\omega) - \sum_{j=1}^{N} C^{ij}(\omega) x^j \right]_+ \right\} < \infty. \quad (18)$$

Then the following statements hold:

(a) *for almost all ω, $\psi_i(x;\omega)$ is a finite-valued and continuous function in $x \in \mathbf{X}$; moreover, for fixed $x^{-i} \in X^{-i}$, $\psi_i(x^i, x^{-i};\omega)$ is convex in $x^i \in X^i$ for almost all ω;*

(b) *for all $x \in \mathbf{X}$, $\psi_i(x;\bullet)$ is an integrable function on Ω;*

(c) *there exists a positive constant $\widehat{\eta}_i$ such that*

$$\mathbb{P}\left\{ \omega \mid |\psi_i(x;\omega)| \leq \widehat{\eta}_i \left(1 + \| x^{-i} \| + \| Q \| \| x^{-i} \|^2 \right) \right\} = 1,$$

for all $x \in \mathbf{X}$;

(d) *if there is a constant $c_i > 0$ such that $|\mathcal{D}_i(\mathcal{Z})| \leq c_i\ essup(|\mathcal{Z}|)$ for all essentially bounded random variables $\mathcal{Z}$, then a constant $\widehat{\eta}'_{i;\lambda_i} > 0$ exists such that for all $x \in \mathbf{X}$,*

$$|\mathcal{R}_{i;\lambda_i}(x)| \leq \widehat{\eta}'_{i;\lambda_i} \left(1 + \| x^{-i} \| + \| Q \| \| x^{-i} \|^2 \right);$$

(e) *for every compact subset $\widehat{\mathbf{X}} \triangleq \prod_{i=1}^{N} \widehat{X}^i$ of $\mathbf{X}$, a constant $\mathrm{Lip}_i > 0$ exists such that for almost all $\omega \in \Omega$ and all $\widehat{x}$ and $\widetilde{x}$ in $\mathbf{X}$,*

$$|\psi_i(\widehat{x};\omega) - \psi_i(\widetilde{x};\omega)| \leq Lip_i \| \widehat{x} - \widetilde{x} \|.$$

Proof **(a)** The two implications (17) imply that the quadratic program defining $\psi_i(x;\omega)$ has a finite optimal solution for almost all $\omega \in \Omega$ and all $x \in \mathbf{X}$. This follows from known QP theory; see e.g. [41, Section 2.2.]. In particular, the first implication ensures the feasibility of the QP and the second implication ensures that the objective is bounded below. These two conditions are necessary and sufficient for the attainment of an optimal solution of the quadratic program defining $\psi_i(x;\omega)$. Hence $\psi_i(x;\omega)$ is

finite for all $x \in \mathbf{X}$ and almost all $\omega \in \Omega$. The convexity of $\psi_i(x^i, x^{-i}; \omega)$ in $x^i \in X^i$ for every $x^{-i} \in X^{-i}$ can be concluded from standard results in SP [71, Section 2.3].

(b) The measurability of $\psi_i(x; \bullet)$ is a well-known result in SP [71, Section 2.3]. Since

$$\psi_i(x; \omega) \;=\; \mathrm{qp}_{\mathrm{opt}}\left(f^i(\omega) + \sum_{j \neq i} G^{ij}(\omega) x^j;\; \xi^i(\omega) - \sum_{j=1}^{N} C^{ij}(\omega) x^j \right), \quad (19)$$

the continuity of $\psi_i(\bullet; \omega)$ on $\mathbf{X}$ follows readily from that of the value function $\mathrm{qp}_{\mathrm{opt}}(f; \xi)$. From part (d) of Lemma 1, we have for some constant $\eta_i > 0$, provided that $\psi_i(x; \omega)$ is finite,

$$\begin{aligned} |\, \psi_i(x; \omega) \,| \;\leq\; \eta \Bigg\{ & \left\| \left(\xi^i(\omega) - \sum_{j=1}^{N} C^{ij}(\omega) x^j \right)_+ \right\| \\ & \times \left[\left\| f^i(\omega) + \sum_{j \neq i} G^{ij}(\omega) x^j \right\| + \| Q \| \left\| \left(\xi^i(\omega) - \sum_{j=1}^{N} C^{ij}(\omega) x^j \right)_+ \right\| \right] \\ & + \| Q \| \left\| f^i(\omega) + \sum_{j \neq i} G^{ij}(\omega) x^j \right\|^2 \Bigg\}. \end{aligned}$$

This shows that $\psi_i(x; \bullet)$ is integrable, establishing part (b).

(c) Under the boundedness assumptions (16) and (18), the existence of the constant $\widehat{\eta}_i > 0$ in part (c) is immediate.

(d) The existence of the scalar $\widehat{\eta}'_{i;\lambda_i}$ in part (d) follows readily from part (c) and the assumption of the deviation measure $\mathcal{D}_i$.

(e) Finally, by part (e) of Lemma 1 and the representation (19) of $\psi_i(x; \omega)$, it follows that there exists a positive constant $\widehat{\mathrm{Lip}}_i$ such that for all $\omega \in \Omega$, all x^{-i}, and all $\widehat{x}^i$ and $\widetilde{x}^{-i}$ as given in part (e) herein, we have

$$|\, \psi_i(\widehat{x}; \omega) - \psi_i(\widetilde{x}; \omega) \,| \;\leq\; \widehat{\mathrm{Lip}}_i \left\| \sum_{j \neq i} G^{ij}(\omega)(\widehat{x}^j - \widetilde{x}^j) + \sum_{j=1}^{N} C^{ij}(\omega)(\widehat{x}^j - \widetilde{x}^j) \right\|$$

Under the boundedness assumption (16), the desired Lipschitz conclusion of part (e) follows readily. □

Under the boundedness assumption (16) on the random coefficient matrices $C^{ij}(\omega)$ and right-hand sides $\xi^i(\omega)$, condition (18) holds if, for instance, $\mathbf{X}$ is bounded or if

$$x \in \mathbf{X} \;\Rightarrow\; \sum_{j=1}^{N} C^{ij}(\omega) x^j \;\geq\; \xi^i(\omega), \quad \text{for almost all } \omega \in \Omega.$$

In turn, if the latter implication holds, then $z^i = 0$ is a feasible solution to the QP defining $\psi_i(x;\omega)$ for almost all $\omega \in \Omega$; hence $\psi_i(x;\omega)$ is upper bounded by zero for all such ω (and lower bounded by a quadratic function of $\|x^{-i}\|$). In the rest of the paper, the assumptions in Proposition 2 will remain in place. Similar to Lemma 1, these assumptions do not yield the boundedness of either the feasible or optimal solutions of the QP that defines the recourse function $\psi_i(x;\omega)$.

4.2 Regularization and differentiability

When the matrix Q is positive definite, the QP (10) has a unique optimal solution, provided that the problem is feasible. This solution, which we denote by $z^{\text{opt}}(f,\xi)$, is related to the dual variable v via the equation $z^{\text{opt}}(f,\xi) = Q^{-1}(-f + D^T v)$ in the constraint of (14). Moreover $z^{\text{opt}}(f,\xi)$ is Lipschitz continuous in the pair $(f;\xi)$ although the dual optimal solution may not be unique in general. Furthermore the optimal objective value $\text{qp}_{\text{opt}}(f;\xi)$ is differentiable in the f-coefficient and its partial gradient with respect to f is given by $\nabla_f \text{qp}_{\text{opt}}(f,\xi) = z^{\text{opt}}(f;\xi)$. Subsequently, we are more interested in the differentiability of $\psi_{\text{opt}}(f;\xi)$ in the ξ-parameter as this is where the players' private variables appear. Noticing that

$$\begin{aligned}\text{qp}_{\text{opt}}(f;\xi) = -\tfrac{1}{2} f^T Q^{-1} f - \underset{v\geq 0}{\text{minimum}} \\ \times \left[\left(-\xi - DQ^{-1}f\right)^T v + \tfrac{1}{2} v^T DQ^{-1}D^T v \right]\end{aligned} \tag{20}$$

we deduce that if D has full row rank so that $DQ^{-1}D^T$ is positive definite, then $\nabla_\xi \text{qp}_{\text{opt}}(f;\xi)$ exists and is equal to $v^{\text{opt}}(f,\xi)$, which is the unique optimal solution (multiplier) of the above dual problem; in this case, this gradient $\nabla_\xi \text{qp}_{\text{opt}}(f;\xi)$ is a Lipschitz function of pair $(f;\xi)$ on the full set Ξ

In the general case when Q is not positive definite or when D is row-rank deficient, the differentiability of $\text{qp}_{\text{opt}}(f;\xi)$ with respect to its arguments (in particular ξ) is in jeopardy. In designing numerical schemes for solving the risk-averse SNEP, particularly iterative best-response type methods, at the very least, we need two conditions on this value function: $\nabla_\xi \text{qp}_{\text{opt}}(f;\xi)$ exists and is Lipschitz continuous in f. In order to achieve both properties, we invoke the dual program (14) that involves the primal variable z and dual variable v jointly and consider the following regularized dual program: for $s > 0$,

$$\begin{aligned}\text{qp}_{s;\text{opt}}(f;\xi) \triangleq \underset{z,v}{\text{maximum}} \xi^T v - \tfrac{1}{2} z^T Qz - \frac{s}{2}\left(z^T z + v^T v\right) \\ \text{subject to} - Qz + D^T v = f, \text{ and } v \geq 0,\end{aligned} \tag{21}$$

where we have regularized both variables z and v. Proposition 3 below summarizes properties of this regularized dual program and its connection to the original QP (10). Alternatively, we may regularize the KKT conditions (11) of the primal quadratic pro-

gram (10), resulting in the following strongly monotone mixed linear complementarity problem:

$$\begin{aligned} 0 &= f + (Q + s\,\mathbf{I})z - D^T v \\ 0 &\le -\xi + Dz + s\,v \;\perp\; v \ge 0, \end{aligned}$$

where $\mathbf{I}$ denotes the identity matrix. The latter set of regularized KKT conditions is equivalent to the optimality conditions of the following convex quadratic program:

$$\underset{v \ge 0}{\text{minimize}} \left\{ \left(-\xi - D(Q + s\mathbf{I})^{-1} f \right)^T v + \tfrac{1}{2} v^T \left[D(Q + s\mathbf{I})^{-1} D^T + s\mathbf{I} \right] v \right\}, \tag{22}$$

which can be compared with (20). Interestingly, the latter (regularized) QP in the v-variable alone does not seem to be immediately related to the jointly regularized dual (21). In what follows, we discuss the latter regularization since it is easier to relate the original objective function $\mathrm{qp}_{\mathrm{opt}}(f;\xi)$ to the regularized objective $\mathrm{qp}_{s;\mathrm{opt}}(f;\xi)$ than to the objective in (22).

Proposition 3 *Let $Q \in \mathbb{R}^{n\times n}$ be a symmetric positive semidefinite matrix and $D \in \mathbb{R}^{m\times n}$ be an arbitrary matrix. For every scalar $s > 0$, the following statements hold:*

(a) *for every f for which (21) is feasible, and for every ξ, this program has a unique pair of optimal solutions, which we denote by $(\widehat{z}^{s;\mathrm{opt}}(f;\xi), \widehat{v}^{s;\mathrm{opt}}(f;\xi))$; moreover, this optimal solution pair is Lipschitz continuous on its domain;*
(b) *for a vector f as given in (a), the optimal objective value $\mathrm{qp}_{s;\mathrm{opt}}(f;\bullet)$ is differentiable with gradient given by $\nabla_\xi \mathrm{qp}_{s;\mathrm{opt}}(f;\xi) = \widehat{v}^{s;\mathrm{opt}}(f;\xi)$; moreover, this gradient is Lipschitz continuous on its domain;*
(c) *for every pair $(f,\xi) \in \Xi$, it holds that $0 \le \mathrm{qp}_{\mathrm{opt}}(f;\xi) - \mathrm{qp}_{s;\mathrm{opt}}(f;\xi) \le \frac{s}{2}\left[c^{Q,D}\,(\|f\| + \|\xi\|)\right]^2$, where $c^{Q,D}$ is the constant in (12).*

Proof **(a)** The existence and uniqueness of an optimal solution to (21) is due to the strong convexity of the regularization term $\frac{s}{2}(z^T z + v^T v)$. The Lipschitz continuity of the optimal solutions as a function of (f,ξ) can be argued as follows. With λ denoting the multiplier of the equality constraint in (21), the KKT conditions of this program can be written as follows:

$$\begin{aligned} 0 &= Qz + sz - Q\lambda \\ 0 &\le -\xi + s\,v + D\lambda \;\perp\; v \ge 0 \\ 0 &= f - D^T v + Qz. \end{aligned}$$

Since this is a mixed linear complementarity problem in the variables (z, v, λ) defined by the positive semidefinite (albeit asymmetric) matrix:

$$\begin{bmatrix} Q + s\mathbf{I} & 0 & -Q \\ 0 & s\mathbf{I} & D \\ Q & -D^T & 0 \end{bmatrix},$$

it follows that the solution map is a polyhedral multifunction of the pair $(f; \xi)$; since the z and v components of any such solution are single-valued functions of $(f; \xi)$, it follows that the solution pair $(\widehat{z}^{s;\mathrm{opt}}(f; \xi), \widehat{v}^{s;\mathrm{opt}}(f; \xi))$ is Lipschitz continuous on the domain of $(f; \xi)$ where (21) is feasible; see [16, Exercise 5.6.4]. This establishes (a).

(b) This part is the consequence of the well-known Danskin's Theorem and part (a).

(c) To prove this part, let (z, v) satisfy (11) and also (12) and let $(\widehat{z}^s, \widehat{v}^s)$ be an optimal pair of (21) corresponding to the same pair (f, ξ). On one hand, we have

$$\begin{aligned} \mathrm{qp}_{s;\mathrm{opt}}(f; \xi) &= \xi^T \widehat{v}^s - \tfrac{1}{2} (\widehat{z}^s)^T Q \widehat{z}^s - \frac{s}{2} \left[(\widehat{z}^s)^T \widehat{z}^s + (\widehat{v}^s)^T \widehat{v}^s \right] \\ &\leq \xi^T \widehat{v}^s - \tfrac{1}{2} (\widehat{z}^s)^T Q \widehat{z}^s \leq \xi^T v - \tfrac{1}{2} z^T Q z = \mathrm{qp}_{\mathrm{opt}}(f; \xi); \end{aligned}$$

on the other hand,

$$\begin{aligned} \mathrm{qp}_{s;\mathrm{opt}}(f; \xi) &\geq \xi^T v - \tfrac{1}{2} v^T Q v - \frac{s}{2} \left[z^T z + v^T v \right] \\ &\geq \mathrm{qp}_{\mathrm{opt}}(f; \xi) - \frac{s}{2} \left[c^{Q,D} \left(\| f \| + \| \xi \| \right) \right]^2, \end{aligned}$$

where the second inequality follows from (12). These two inequalities establish part (c). □

4.3 Quantile-based deviation measures

To motivate the class of mean-deviation functions $\mathcal{R}_{i;\lambda_i}(x)$ for which the risk-averse SNEP can be treated constructively, consider a quantile (or CVAR)-based deviation measure $\mathcal{D}_\gamma^{\mathrm{QD}}(\mathcal{Z})$ that leads to the mean-deviation function $\mathcal{R}_{\lambda,\gamma}^{\mathrm{QD}}(x) \triangleq \mathbb{E}[\psi(x; \widetilde{\omega})] + \lambda \mathcal{D}_\gamma^{\mathrm{QD}}[\psi(x; \widetilde{\omega})]$ having the desired convexity property in the players' individual variables. Throughout this subsection, we drop the index i pertaining to the player's label and present a general discussion without referring to a game. In describing the results below about mean-quantile deviations, we have borrowed significantly from the references [64], [71, Chapter 6] and [74]. For a scalar t, the notation $t_\pm$ denotes the nonnegative and nonpositive part of t, respectively; i.e., $t_+ \triangleq \max(t, 0)$ and $t_- \triangleq \max(-t, 0)$. This notation extends to vectors in terms of their components.

Specifically, for $\gamma \in [0, 1]$, let [52,53]

$$\begin{aligned}\mathcal{D}_\gamma^{\mathrm{QD}}(\mathcal{Z}) &\triangleq \mathbb{E}\left\{\gamma\,\left[\,\mathcal{Z}-\kappa_\gamma(\mathcal{Z})\,\right]_+ + (\,1-\gamma\,)\,\left[\,\mathcal{Z}-\kappa_\gamma(\mathcal{Z})\,\right]_-\right\}\\ &= \underset{t\in\mathbb{R}}{\text{minimum}}\,\mathbb{E}\left[\gamma\,(\,\mathcal{Z}-t\,)_+ + (\,1-\gamma\,)\,(\,\mathcal{Z}-t\,)_-\right],\end{aligned}$$

with the minimizer being the γ-quantile $\kappa_\gamma(\mathcal{Z})$ of the random variable $\mathcal{Z}$ for $\gamma \in (0, 1)$. Note that if $\mathcal{Z}$ is essentially bounded, then $\mathcal{D}_0^{\mathrm{QD}}(\mathcal{Z}) = \mathcal{D}_1^{\mathrm{QD}}(\mathcal{Z}) = 0$. With the deviation measure $\mathcal{D}_\gamma^{\mathrm{QD}}(\mathcal{Z})$, we have

$$\begin{aligned}\mathcal{R}_{\lambda,\gamma}^{\mathrm{QD}}(\mathcal{Z}) &= \mathbb{E}\,[\mathcal{Z}] + \lambda\,\mathcal{D}_\gamma^{\mathrm{QD}}(\mathcal{Z})\\ &= \underset{t\in\mathbb{R}}{\text{minimum}}\,\mathbb{E}\left[\,\mathcal{Z}+\lambda\,\gamma\,(\,\mathcal{Z}-t\,)_+ + \lambda\,(\,1-\gamma\,)\,(\,\mathcal{Z}-t\,)_-\,\right]\\ &= \underset{t\in\mathbb{R}}{\text{minimum}}\,\mathbb{E}\left[\underbrace{t + (\,1-\lambda+\lambda\,\gamma\,)\,(\,\mathcal{Z}-t\,) + \lambda\,(\,\mathcal{Z}-t\,)_+}_{H_\lambda^{\mathrm{QD}}(\mathcal{Z};t;\gamma)}\right].\end{aligned} \tag{23}$$

We note that for all $\gamma \in (0, 1)$, we have $|\kappa_\gamma(\mathcal{Z})| \le \text{essup}(|\mathcal{Z}|)$. Indeed, this follows from the fact that $\kappa_\gamma(\mathcal{Z})$ is the scalar t satisfying $\mathbb{P}\{\mathcal{Z} < t\} \le \gamma$ and $\mathbb{P}\{\mathcal{Z} \le t\} \ge \gamma$ and the expression

$$\text{essup}(|\mathcal{Z}|) \;=\; \inf\,\{\,\alpha \;\mid\; \mathbb{P}\{|\mathcal{Z}| \le \alpha\} = 1\,\}\,.$$

The function $H_\lambda^{\mathrm{QD}}(z; t; \gamma)$ has the following favorable properties for (fixed) scalars λ and (possibly varying) γ, both restricted to be in [0, 1]:

- it is Lipschitz continuous in its arguments, i.e., in $(z; t; \gamma)$;
- it is strictly increasing in z for fixed (t, γ) unless $1 - \lambda + \lambda\gamma = 0$ (this is the exceptional case for $\lambda \in [0, 1]$ that we will exclude below);
- it is convex and piecewise linear in (z, t) jointly for fixed γ; and lastly
- it is linear in γ for fixed (z, t).

Since $H_\lambda^{\mathrm{QD}}(\bullet; t; \gamma)$ is strictly increasing (except in the excluded case where $1 - \lambda + \lambda\gamma = 0$) and provided that $\psi(\bullet; \omega)$ is convex for almost all $\omega \in \Omega$, we deduce that the composite function $H_\lambda^{\mathrm{QD}}(\psi(x; \omega); t; \gamma)$ is convex in x. Thus, if $|\psi(x; \omega)| \le \eta(1+\|x\|)$ for some constant $\eta > 0$ and almost all ω [this is guaranteed by Proposition 2 under the relatively complete recourse assumption], then

$$\left|\,\mathcal{R}_{\lambda,\gamma}^{\mathrm{QD}}(x)\,\right| \;\le\; (\,\eta + 2\,\lambda\,)\,(\,1 + \|x\|\,). \tag{24}$$

We mention two interesting classes of deviation measures derived from the above quantile-based deviation; namely,

- absolute semi-deviation (ASD):

$$\mathcal{D}^{\mathrm{ASD}}(\mathcal{Z}) \triangleq \mathbb{E}\,[\mathcal{Z} - \mathbb{E}\mathcal{Z}]_+ = \underset{\gamma\in[0,1]}{\text{maximum}}\, \mathcal{D}_\gamma^{\mathrm{QD}}(\mathcal{Z}),$$

  where the second equality is proved in [51]; and
- absolute deviation (AD):

$$\mathcal{D}^{\mathrm{AD}}(\mathcal{Z}) \triangleq \mathbb{E}\,|\,\mathcal{Z} - \mathbb{E}\mathcal{Z}\,| = 2\,\mathbb{E}\,[\,\mathcal{Z} - \mathbb{E}\mathcal{Z}\,]_+ .$$

Generalizing the specific function $H_\lambda^{\mathrm{QD}}(z;t;\gamma)$ defined in (23), we introduce an abstract function $H_\lambda(z;t;\gamma)$ based on which we define a max-min (motivated by the ASD) mean-deviation function and its composition with a recourse function $\psi(x;\omega)$:

$$\varphi_\lambda(\mathcal{Z}) \triangleq \underset{\gamma\in\Gamma}{\text{maximum}}\ \underset{t\in\mathbb{R}}{\text{minimum}}\ \mathbb{E}\left[\,H_\lambda(\mathcal{Z};t;\gamma)\,\right] \text{ and } \widehat{\psi}_\lambda(x) \triangleq \varphi_\lambda(\psi(x;\widetilde{\omega})),$$

where Γ is a compact interval on the real line.

The abstract function $H_\lambda(z;t;\gamma)$ accommodates a mean-CVAR objective as follows: for $\gamma \in (0,1)$,

$$\begin{aligned}\widehat{\psi}_\lambda^{\mathrm{CVAR}}(x) &\triangleq \mathbb{E}\,[\psi(x;\widetilde{\omega})] + \lambda\,\mathrm{CVAR}_\gamma\,[\psi(x;\widetilde{\omega})]\\ &= \mathbb{E}\,[\psi(x;\widetilde{\omega})] + \lambda\,\underset{t\in\mathbb{R}}{\text{minimum}}\left\{t + \frac{1}{1-\gamma}\,\mathbb{E}\,[\psi(x;\widetilde{\omega}) - t]_+\right\}\\ &= \underset{t\in\mathbb{R}}{\text{minimum}}\left[\mathbb{E}\,[\psi(x;\widetilde{\omega})] + \lambda\left\{t + \frac{1}{1-\gamma}\,\mathbb{E}\,[\psi(x;\widetilde{\omega}) - t]_+\right\}\right],\end{aligned}$$

which has $H_\lambda^{\mathrm{CVAR}}(z;t;\gamma) = z + \lambda\left\{t + \dfrac{1}{1-\gamma}\,[z-t]_+\right\}$; this is a minor variation of the function $H_\lambda^{\mathrm{QD}}(z;t;\gamma)$. The set Γ is a singleton in the mean-CVAR function $\widehat{\psi}_\lambda^{\mathrm{CVAR}}(x)$. While $H_\lambda^{\mathrm{CVAR}}(z;t;\gamma)$ is not linear in γ (in fact it is convex), but there is no maximization in the γ-variable for this function.

We introduce two properties of the function H_λ that generalize those of the special cases of H_λ^{QD} and $H_\lambda^{\mathrm{CVAR}}$; the first property (cf. the same properties of H_λ^{QD}) ensures the satisfaction of the required convexity of the composite function $\widehat{\psi}_\lambda$ in a game-theoretic setting:

Assumption (H1) H_λ is Lipschitz continuous and strictly increasing in z for fixed (t,γ); convex and piecewise linear in (z,t) jointly for fixed γ, and linear in γ for fixed (z,t). □

The second property is motivated by the linear growth property (24) of $\left|\mathcal{R}_{\lambda,\gamma}^{\mathrm{QD}}(x)\right|$ in terms of $\|x\|$:

Assumption (H2) There exists a constant $\eta_H > 0$ such that $|H_\lambda(z;t;\gamma)| \le \eta_H(1 + \|z\|)$ for all $(t;\gamma) \in \mathbb{R}\times\Gamma$ and all z of interest. □

4.4 The reformulated mean-deviation composite game $\mathcal{G}^{\text{rfMD}}$

Based on an abstract mean-deviation function $H_{i;\lambda_i}(z; t_i; \gamma_i)$ for player i that satisfies assumptions (H1) and (H2), the problem (8) becomes:

$$\underset{x^i \in X^i}{\text{minimize}} \left\{ \theta_i(x^i, x^{-i}) + \max_{\gamma_i \in \Gamma_i} \min_{t_i \in \mathbb{R}} \underbrace{\mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x^i, x^{-i}; \widetilde{\omega}); t_i; \gamma_i) \right]}_{h_{i;\lambda_i}(x^i, x^{-i}; t_i; \gamma_i)} \right\}. \quad (25)$$

As mentioned above, the function H_λ^{CVAR} is not linear in γ and therefore does not satisfy the linearity requirement in γ in Assumption (H1). However, the development below is applicable to this function too because the interval Γ is a singleton; it suffices to identify the two end points of this interval to coincide. Specifically, in the minmax optimization problems (26) of the reformulated game $\mathcal{G}^{\text{rfMD}}$ below, we have $\underline{\gamma}_i = \overline{\gamma}_i$ for all $i = 1, \ldots, N$, so that the two-term max becomes the single term $\mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x^i, x^{-i}; \widetilde{\omega}); t_i; \widehat{\gamma}_i) \right]$, where $\widehat{\gamma}_i$ is the single element in the interval Γ_i.

Under assumption (H1), the function $(t_i; \gamma_i) \mapsto h_{i;\lambda_i}(x; t_i; \gamma_i)$ is of the convex-linear type for fixed x; i.e., $h_{i;\lambda_i}(x; \bullet; \gamma_i)$ is convex and $h_{i;\lambda_i}(x; t_i; \bullet)$ is linear. Thus, problem (25) is of the minmax-min kind. Motivated by the bound of the quantile: $|\kappa_\gamma(\mathcal{Z})| \leq \text{essup}(|\mathcal{Z}|)$, we introduce the following assumption.

Assumption (H3) for each player $i = 1, \ldots, N$, there exists a compact interval $\mathcal{T}_i$ such that

$$\underset{t_i \in \mathbb{R}}{\text{minimum}}\, h_{i;\lambda_i}(x; t_i; \gamma_i) = \underset{t_i \in \mathcal{T}_i}{\text{minimum}}\, h_{i;\lambda_i}(x; t_i; \gamma_i)$$

for all $x \in \mathbf{X}$ and every $\gamma_i \in \Gamma_i$. □

Assumption (H3) is satisfied if for example when the minimizer in the t-variable is a quantile of the random variable $\psi_i(x; \widetilde{\omega})$ such that the essential supremum of $|\psi_i(x; \widetilde{\omega})|$ is bounded by a multiple of $(1 + \|x\|)$. In general, under this assumption, the min-max-min problem (25) is equivalent to a minmax problem:

$$\underset{x^i \in X^i,\, t_i \in \mathcal{T}_i}{\text{minimize}} \left\{ \theta_i(x^i, x^{-i}) + \max_{\gamma_i \in \Gamma_i} h_{i;\lambda_i}(x^i, x^{-i}; t_i; \gamma_i) \right\},$$

by simply interchanging the max-min in the γ_i and t_i variables, respectively, to a min-max in the interchanged variables. [The convexity-linearity of $H_{i;\lambda_i}(\psi_i(x; \widetilde{\omega}); \bullet; \bullet)$ justifies this interchange.] Furthermore, since $h_{i;\lambda_i}(x^i, x^{-i}; t_i; \gamma_i)$ is linear in γ_i, letting $\underline{\gamma}_i$ and $\overline{\gamma}_i$ be the two (finite) endpoints of the interval Γ_i, we have

$$\underset{\gamma_i \in \Gamma_i}{\text{maximum}}\, h_{i;\lambda_i}(x^i, x^{-i}; t_i; \gamma_i) = \max\left(h_{i;\lambda_i}(x^i, x^{-i}; t_i; \underline{\gamma}_i), h_{i;\lambda_i}(x^i, x^{-i}; t_i; \overline{\gamma}_i) \right).$$

Thus, problem (25) simplifies to

$$\underset{x^i \in X^i,\, t_i \in \mathcal{T}_i}{\text{minimize}} \left\{ \theta_i(x^i, x^{-i}) + \max\left(h_{i;\lambda_i}(x^i, x^{-i}; t_i; \underline{\gamma}_i),\ h_{i;\lambda_i}(x^i, x^{-i}; t_i; \overline{\gamma}_i) \right) \right\},$$

which, after substitution of $h_{i;\lambda_i}(x^i, x^{-i}; t_i; \underline{\gamma}_i)$ and $h_{i;\lambda_i}(x^i, x^{-i}; t_i; \overline{\gamma}_i)$ results in the following equivalent formulation of player i's optimization problem:

$$\begin{aligned} \underset{x^i \in X^i,\, t_i \in \mathcal{T}_i}{\text{minimize}} & \left\{ \theta_i(x^i, x^{-i}) \right. \\ & \left. + \max\left(\mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x^i, x^{-i}; \widetilde{\omega}); t_i; \underline{\gamma}_i) \right],\ \mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x^i, x^{-i}; \widetilde{\omega}); t_i; \overline{\gamma}_i) \right] \right) \right\}. \end{aligned} \tag{26}$$

Formally, the game $\mathcal{G}^{\text{rfMD}}$ consists of the above player optimization problems (26) for $i = 1, \ldots, N$. This formulation is the starting point where the two-stage SNEP with risk-averse players is amenable to constructive analysis and algorithmic design. Beginning in the next section, we discuss solution approaches for approximating/solving this game, as a by-product of which existence results of a Nash solution to the original risk-averse game can be established constructively. Here, we remind the reader that the linear growth assumption (H2) and the same property of the recourse function $\psi_i(x; \omega)$ ensures the continuity of the objective function of (26) in x^i.

Having resolved one technical challenge associated with the min-max-min formulation, we are still faced with several other challenges in the design of computational algorithms for solving the game $\mathcal{G}^{\text{rfMD}}$. Specifically, these technical challenges are summarized below:

- the value function $\psi_i(x; \omega)$ involves an inner minimization possibly with non-unique minimizers and the coupling of the players' variables in this non-differentiable function;
- the possible non-uniqueness of the minimizing t_i-variable and the lack of strong convexity in this variable that could affect the convergence of an overall iterative algorithm;
- approximation of the expectation operator that lies at the heart of the solution of a stochastic program;
- the nonsmoothness of the function $H_{i;\lambda_i}(z; t_i; \gamma_i)$ (such as the plus function $[\bullet]_+$ in both the mean-quantile deviation function H_λ^{QD} and the mean-CVAR function H_λ^{CVAR}) and the nondifferentiability of the 2-term pointwise max function

$$\begin{aligned} \widehat{h}_{i,\lambda_i}(x; t_i) \triangleq \max\Big(& \mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x; \widetilde{\omega}); t_i; \underline{\gamma}_i) \right], \\ & \mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x; \widetilde{\omega}); t_i; \overline{\gamma}_i) \right] \Big). \end{aligned} \tag{27}$$

Each of the above features by itself is a non-trivial issue; having all of them together makes the solution of the game $\mathcal{G}^{\text{rfMD}}$ highly complicated. Figure 1 provides a schematic of how risk-neutral and robust Nash games may be derived from the reformulated mean deviation composite Nash game.

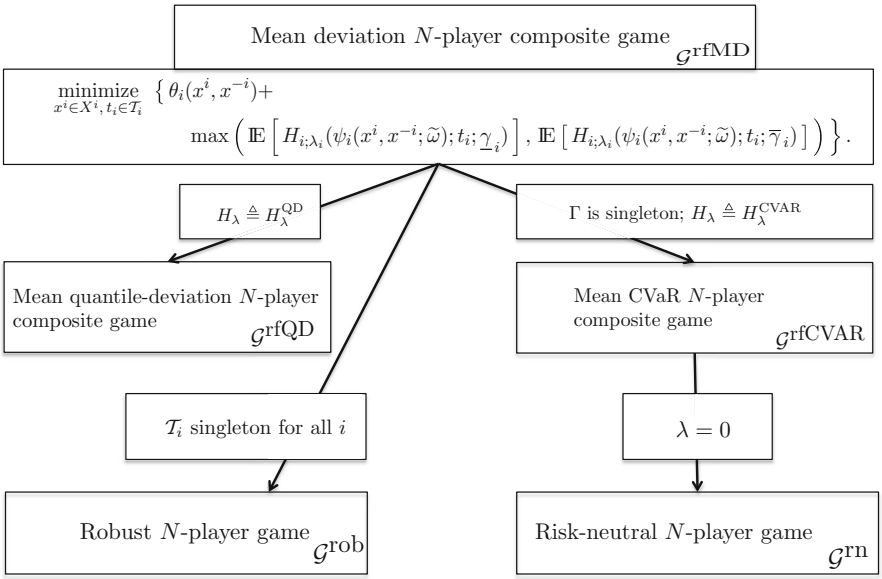

Fig. 1 Relationships among the various forms of risk-averse Nash games

We close this introduction of the game $\mathcal{G}^{\mathrm{rfMD}}$ with two remarks. One, each optimization problem (26) is a minmax stochastic program, with the inner max term being very simple; moreover, referring to the problem (5), we can see that mathematically the former is a special case of the latter although they each have their own modeling sources. Two, with the preparatory efforts to ensure the convexity in player i's overall objective in (26), the existence of a Nash equilibrium solution to the game $\mathcal{G}^{\mathrm{rfMD}}$ follows from classic results in non-cooperative game theory provided that each set X^i is bounded. When these sets are unbounded, one avenue for deriving existing statements for stochastic Nash games arises from combining Lebesgue convergence theorems with coercivity arguments (see [55]). However, this approach requires further study; in this paper, we focus on the computation/approximation of such an NE, a key challenge associated with this game.

4.5 Families of games for solution

Having arrived at the reformulated game $\mathcal{G}^{\mathrm{rfMD}}$ defined using the family $\left\{ H_{i;\lambda_i} \right\}_{i=1}^N$ of abstract mean-deviation functions satisfying assumptions (H1) and (H2), we summarize the remaining sections that address the solution of this game. Section 5 focuses on an important special case, denoted $\mathcal{G}^{\mathrm{prMD}}$ and defined by the optimization problems (28), where the players' recourse functions are all private, i.e., these functions do not contain the rivals' variables; both a deterministic and a sampled best-response algorithm are presented and in each instance, convergence is established. The former algorithm involves solving a sequence of stochastic optimization problems (31) and the

latter involves solving a sequence of sampled Nash games $\left\{\mathcal{G}^{\,\mathrm{prMD};L_\nu}\right\}_{\nu=1}^{\infty}$. Section 6 addresses the reformulated game $\mathcal{G}^{\mathrm{rfMD}}$ with coupling variables in the players' recourse functions. A sequence of smoothed and regularized games $\left\{\mathcal{G}_{s_k}^{\mathrm{srMD}}\right\}$ corresponding to a prescribed sequnece of positive scalars $\{s_k\}$ tending to zero is constructed and the solution of each game $\mathcal{G}_s^{\mathrm{srMD}}$ is discussed. A roadmap of the details of the overall solution process is presented ahead of the technical discussion.

5 A special case: separable recourse functions

As a first step in addressing the challenges of the game $\mathcal{G}^{\,\mathrm{rfMD}}$, we consider a special case where no additional maneuvering is needed for its solution. Indeed, existing game-theoretic algorithms combined with stochastic programming methods (suitably extended) can be directly applied for computing a Nash equilibrium. In this case, each player's recourse is *private*, i.e., the function $\psi_i(x;\omega)$ does not contain the rivals' variables x^{-i} in both the objective function and constraint of (9). As a matter of fact, the development below is not restricted to quadratic recourse functions as long as there are no coupling of the players' variables in these functions. For concreteness, we continue to focus on quadratic recourse functions as given by

$$\begin{aligned}\psi_i(x^i;\omega) \triangleq \underset{z^i}{\text{minimum}} \quad & \left(f^{\,i}(\omega)\right)^T z^i + \tfrac{1}{2}\,(z^i)^T Q^i z^i \\ \text{subject to} \quad & C^{ii}(\omega)x^i + D^i z^i \;\geq\; \xi^{\,i}(\omega)\end{aligned} \tag{28}$$

We term the resulting game as one with private recourse and denote it by $\mathcal{G}^{\,\mathrm{prMD}}$; this game consists of the following player optimization problems:

$$\left\{\underset{x^i\in X^i,\,t_i\in\mathcal{T}_i}{\text{minimize}}\;\left\{\theta_i(x^i,x^{-i})+\varphi_i(x^i;t_i)\right\}\right\}_{i=1}^{N}, \tag{29}$$

where the function

$$\varphi_i(x^i;t_i) \;\triangleq\; \max\left(\mathbb{E}\left[\,H_{i;\lambda_i}(\psi_i(x^i;\widetilde{\omega});t_i;\underline{\gamma}_i)\,\right],\;\mathbb{E}\left[\,H_{i;\lambda_i}(\psi_i(x^i;\widetilde{\omega});t_i;\overline{\gamma}_i)\,\right]\right) \tag{30}$$

is private to player i. [Remark: we change the notation from $\widehat{h}_{i,\lambda_i}(x,t_i)$ in (27) to $\varphi_i(x^i;t_i)$ to emphasize the absence of the variable x^{-i} in the recourse term.] Each of the problems in (29) is a simplified form of the problem (26). The function $\varphi_i(x^i;t_i)$ remains non-differentiable but is jointly convex in $(x^i;t_i)$. Taking advantage of the separability in the players' private variables in this function, we propose the following direct adaption of a deterministic best-response algorithm for computing an equilibrium of the game $\mathcal{G}^{\,\mathrm{prMD}}$.

Deterministic best-response algorithm for $\mathcal{G}^{\text{prMD}}$. At iteration $\nu = 0, 1, 2, \ldots$, given an iterate $x^\nu \triangleq \left(x^{\nu;i}\right)_{i=1}^N$ in $\mathbf{X}$ let each

$$(x^{\nu+1;i}, t_i^{\nu+1}) \in \underset{x^i \in X^i,\, t_i \in \mathcal{T}_i}{\operatorname{argmin}} \left\{ \theta_i(x^i, x^{\nu;-i}) + \varphi_i(x^i; t_i) + \tfrac{1}{2} \| x^i - x^{\nu;i} \|^2 \right\}; \tag{31}$$

$$i = 1, \ldots, N.$$

Iterate until a prescribed termination criterion is satisfied. □

The termination criterion (for example, the difference between two successive iterates is within a prescribed tolerance) is used to stop the algorithm in a finite of iterations. As with a typical analysis of all deterministic iterative algorithms, the goal is to establish a desirable solution property of a limit point of an infinite sequence of iterates produced by the algorithm. Note that each optimization subproblem (31) is a convex stochastic program with convex quadratic recourse; methods from SP can therefore be borrowed to solved these subproblems. Yet, the two-term max operator is an unusual feature of these subproblems. Thus, care is needed to apply the SP methodology to them. Here, we do not concern ourselves with the details of how these subproblems can be solved (see [74]); instead we proceed to present a convergence result of the algorithm, by borrowing from the recent advances on the best-response algorithm, see e.g., [17, Section 12.6]. Theorem 4 below gives a complete convergence result for the above algorithm, under the assumption that the family of objective functions $\{\theta_i\}_{i=1}^N$ satisfies the following

Assumption (DD) each function $\theta_i(x)$ is continuous; $\nabla_{x^i}\theta_i(x)$ exists and is continuously differentiable on $\mathbf{X}$ with derivatives satisfying:

- $\gamma_{ii} \triangleq \inf_{x \in \mathbf{X}}$ smallest eigenvalue of $\nabla^2_{x^i}\theta_i(x) \in (0, \infty)$, for all $i = 1, \ldots, N$;
- $\gamma_{ij} \triangleq \sup_{x \in \mathbf{X}} \left\| \nabla^2_{x^j x^i}\theta_i(x) \right\| < \infty$, for all $i \neq j = 1, \ldots, N$; and
- the matrix $\Upsilon \triangleq \left[\gamma_{ij}\right]_{i,j=1}^N$ is such that its comparison matrix is of class P [12].

□

The latter P-condition is equivalent to the existence of positive constants $d_i > 0$ for $i = 1, \ldots, N$ such that

$$\gamma_{ii}\, d_i > \sum_{j \neq i} \gamma_{ij}\, d_j, \ \forall\, i = 1, \ldots, N.$$

This condition is a generalized diagonal dominance property that extends the familiar notion of diagonal dominance with the constants d_i all equal to unity; thus the acronym (DD) of the assumption. Assumption (DD) is the basic condition used in [17] to establish the convergence of the best-response algorithm for a deterministic Nash equilibrium problem. In what follows, we employ the same assumption for the proof of convergence of the adaptation of this algorithm for solving the SNEP. Under

this assumption, it follows that the objective function in (31) is strongly convex in x^i for fixed (x^{-i}, t_i). Since the constraint set of this problem is the Cartesian product of the closed convex set X^i and the compact interval $\mathcal{T}_i$, it follows that $(x^{\nu+1;i}; t_i^{\nu+1})$ is well defined (because the objective function of (31) is coercive on the feasible set) and $x^{\nu+1;i}$ is unique but $t_i^{\nu+1}$ is not necessarily so. The non-uniqueness of the optimal t-variable in (31) is a slight departure from the existing convergence theory of best-response algorithms which typically requires the strong convexity and thus uniqueness of the optimal solutions of the subproblems. The convergence of the sequence $\left\{((x^{\nu+1;i}; t_i^{\nu+1})\right\}$ of best-response iterates is stated and proved in the result below.

Theorem 4 *Suppose that each $\varphi_i(x^i; t_i)$ is a continuous and convex function on $X^i \times \mathcal{T}_i$. Under assumption (DD), the following two statements hold:*

(a) *the best-response algorithm produces a well-defined sequence $\{(x^\nu; t^\nu)\}$ where $x^\nu \triangleq \left(x^{\nu;i}\right)_{i=1}^N$ and $t^\nu \triangleq \left(t_i^\nu\right)_{i=1}^N$ with each $x^{\nu+1}$ being uniquely defined;*

(b) *the sequence $\{x^\nu\}$ converges to a vector, say x^∞; if $t^\infty \triangleq \left(t_i^\infty\right)_{i=1}^N$ is any accumulation point of any such sequence $\{t^\nu\}$, then the pair $(x^\infty; t^\infty)$ is a Nash equilibrium solution of the game $\mathcal{G}^{\text{prMD}}$.*

Proof We omit (a) and proceed to prove the convergence assertion in part (b). For each (ν, i), we have

$$\begin{aligned}&(x^i - x^{\nu+1;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) + x^{\nu+1;i} - x^{\nu;i}\right]\\ &\quad + \varphi_i(x^i; t_i) - \varphi_i(x^{\nu+1;i}; t_i^{\nu+1}) \;\geq\; 0,\end{aligned} \tag{32}$$

for all $(x^i; t_i) \in X^i \times \mathcal{T}_i$. Thus,

$$\begin{aligned}&(x^{\nu;i} - x^{\nu+1;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) + x^{\nu+1;i} - x^{\nu;i}\right]\\ &\quad + \varphi_i(x^{\nu;i}; t_i^\nu) - \varphi_i(x^{\nu+1;i}; t_i^{\nu+1}) \;\geq\; 0.\end{aligned}$$

Similarly, we also have

$$\begin{aligned}&(x^{\nu+1;i} - x^{\nu;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu;i}, x^{\nu-1;-i}) + x^{\nu;i} - x^{\nu-1;i}\right] + \varphi_i(x^{\nu+1;i}; t_i^{\nu+1})\\ &\quad - \varphi_i(x^{\nu;i}; t_i^\nu) \;\geq\; 0.\end{aligned}$$

Adding the two inequalities, we deduce

$$\begin{aligned}&(x^{\nu+1;i} - x^{\nu;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) - \nabla_{x^i}\theta_i(x^{\nu;i}, x^{\nu-1;-i}) + x^{\nu+1;i} - x^{\nu;i}\right]\\ &\quad \leq (x^{\nu+1;i} - x^{\nu;i})^T (x^{\nu;i} - x^{\nu-1;i}).\end{aligned}$$

At this point, we can apply a standard argument to establish a contraction property of the sequence $\{x^\nu\}$; cf. e.g. the proof of [18, Theorem 3]. Specifically, it follows by applying the mean-value theorem to the above inequality that

$$(1+\gamma_{ii})\,\|x^{\nu+1;i}-x^{\nu;i}\|^2 \;\le\; \left[\,\|x^{\nu;i}-x^{\nu-1;i}\| + \sum_{j\neq i}\gamma_{ij}\,\|x^{\nu;j}-x^{\nu-1;j}\|\,\right] \|x^{\nu+1;i}-x^{\nu;i}\|$$

holds for all $i = 1, \ldots, N$. Concatenating these N inequalities, we obtain

$$\begin{pmatrix} \|x^{\nu+1;1}-x^{\nu;1}\| \\ \vdots \\ \|x^{\nu+1;N}-x^{\nu;N}\| \end{pmatrix} \le \widehat{\Upsilon} \begin{pmatrix} \|x^{\nu;1}-x^{\nu-1;1}\| \\ \vdots \\ \|x^{\nu;N}-x^{\nu-1;N}\| \end{pmatrix},$$

where $\widehat{\Upsilon}$ is the nonnegative matrix

$$\begin{bmatrix} \frac{1}{1+\gamma_{11}} & \frac{\gamma_{12}}{1+\gamma_{11}} & \frac{\gamma_{13}}{1+\gamma_{11}} & \cdots & \frac{\gamma_{1N}}{1+\gamma_{11}} \\ \frac{\gamma_{21}}{1+\gamma_{22}} & \frac{1}{1+\gamma_{22}} & \frac{\gamma_{23}}{1+\gamma_{22}} & \cdots & \frac{\gamma_{2N}}{1+\gamma_{NN}} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{\gamma_{N1}}{1+\gamma_{NN}} & \frac{\gamma_{N2}}{1+\gamma_{NN}} & \cdots & \frac{\gamma_{NN-1}}{1+\gamma_{NN}} & \frac{1}{1+\gamma_{NN}} \end{bmatrix}.$$

The P-property of the comparison matrix of Υ implies that $\widehat{\Upsilon}$ has spectral radius less than unity. Thus there exists a scalar $\delta \in (0, 1)$ and a monotonic norm $|||\bullet|||$ such that

$$\left|\left|\left| \begin{pmatrix} \|x^{\nu+1;1}-x^{\nu;1}\| \\ \vdots \\ \|x^{\nu+1;N}-x^{\nu;N}\| \end{pmatrix} \right|\right|\right| \le \delta \left|\left|\left| \begin{pmatrix} \|x^{\nu;1}-x^{\nu-1;1}\| \\ \vdots \\ \|x^{\nu;N}-x^{\nu-1;N}\| \end{pmatrix} \right|\right|\right|.$$

This shows that the sequence $\{x^\nu\}$ contracts and thus converges to a limit x^∞. The sequence $\{t^\nu\} \subset \prod_{i=1}^N \mathcal{T}_i$ is bounded, and thus has accumulation point(s). Let t^∞ be the limit of a convergent subsequence $\{t^{\nu+1}\}_{\nu\in\kappa}$. Since (32) holds for all pairs (ν, i), passing to the limit as $\nu(\in \kappa) \to \infty$, we deduce

$$(x^i - x^{\infty;i})^T \nabla_{x^i}\theta_i(x^{\infty;i}, x^{\infty;-i}) + \varphi_i(x^i; t_i) - \varphi_i(x^{\infty;i}; t_i^\infty) \;\ge\; 0,$$

for all $(x^i; t_i) \in X^i \times \mathcal{T}_i$. This inequality is necessary and sufficient for the pair $(x^\infty; t^\infty)$ to be a NE of the game $\mathcal{G}^{\text{prMD}}$ defined by the optimization problems (29). □

It is interesting to note that in this case of private recourse functions, the main convergence condition (DD) is on the deterministic first-stage objective functions only. Another noteworthy remark is that under the (DD) condition, the x-part of any NE of the game $\mathcal{G}^{\text{prMD}}$ must be unique while the t-part is not guaranteed to be unique.

The uniqueness claim of the x-part can be proved by following a similar argument as the contraction proof in the theorem. Details are not repeated.

5.1 A sequential sampled solution approach

The convergence result (4) is deterministic in nature; it assumes that an exact, more importantly, deterministic solution to each subproblem can be computed. This assumption, however, is practically difficult to satisfy except in the case where the sample space Ω is finite, say $\Omega = \{\omega^1, \ldots, \omega^K\}$ for some positive integer K, and the respective probabilities p_ℓ for $\ell = 1, \ldots, K$ of the samples are available. In this case, player i's subproblem in (31) is

$$\min_{x^i \in X^i,\, t_i \in \mathcal{T}_i} \left\{ \theta_i(x^i, x^{\nu;-i}) + \max \left(\sum_{\ell=1}^{K} p_\ell \, H_{i;\lambda_i}(\psi_i(x^i; \omega^\ell); t_i; \underline{\gamma}_i), \right.\right.$$
$$\left.\left. \sum_{j=1}^{K} p_\ell \, H_{i;\lambda_i}(\psi_i(x^i; \omega^\ell); t_i; \overline{\gamma}_i) \right) \right\}.$$

These scenario-based subproblems also arise when one employs a sampling approach when the space Ω is a continuum. In this case, the family $\{\omega^1, \ldots, \omega^K\}$ consists of K independently and identically distributed samples of the random variable $\widetilde{\omega}$ and $\{p_1, \ldots, p_K\}$ are identically $1/K$; moreover, the integer K and the families of samples and scalars may vary from one iteration ν to the next.

In this subsection and the next, we expand on the above idea and present two sampling based algorithms for solving the game $\mathcal{G}^{\text{prMD}}$ and address their convergence. We first present an algorithm that solves a sequence of sampled games obtained by taking increasingly many samples as the iteration proceeds. Throughout we do not address the details of the sampling process; instead we simply let $\{\omega^1, \ldots, \omega^L\}$ be L independently and identically distributed samples of the random variable $\widetilde{\omega}$ such that the recourse function $\psi_i(x^i; \omega^j)$ is finite for all $x^i \in X^i$ and all $j = 1, \ldots, L$. [Admittedly, under the "almost all" assumptions of Proposition 2 that ensure the finiteness of $\psi_i(x^i; \omega)$ for almost all (and not necessarily for all) realizations ω of the uncertainty, the sampling process is not a trivial task. We make two remarks regarding this issue. First, minimally addressed to date, this issue exists in all sampling methods for solving two-stage stochastic programs where uncertainty arises in the second stage not all of whose realizations lead to finiteness of the recourse function. Second, there is a possible way to remedy this difficulty by applying regularization plus a least constraint violation idea wherein the objective of the recourse function is strongly convexified and the second-stage constraint is modified using a minimum residual of the constraint violation. The latter solution approach is new, even in the context of standard two-stage SPs. Nevertheless to analyze the approach is beyond the scope of this paper. For these reasons, we leave aside the issue of the sampling process throughout the remainder of this paper.]

We denote by $\mathcal{G}^{\mathrm{prMD};L}$ the game composed of the following optimization problems:

$$\left\{ \theta_i \left(\underbrace{x^i, x^{-i}}_{\text{sampled game variable}} \right) + \frac{1}{L} \max \left(\sum_{\ell=1}^{L} H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell); t_i; \underline{\gamma}_i), \right.\right.$$
$$\left.\left. \sum_{\ell=1}^{L} H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell); t_i; \overline{\gamma}_i) \right) \right\}_{i=1}^{N},$$

where we have approximated the two expectations in the function $\varphi_i(x^i; t_i)$ (cf. (30)) by two finite sums derived from the given samples and have employed equal probabilities for each of the samples. Omitting the details of how the sampled game $\mathcal{G}^{\mathrm{prMD};L}$ is solved, we formally present the following:

Sampling based algorithm for $\mathcal{G}^{\mathrm{prMD}}$. At iteration $\nu = 0, 1, 2, \ldots$, given the samples $\{\omega_\ell^\nu\}_{\ell=1}^{L_\nu}$ in Ω, solve the sampled game $\mathcal{G}^{\mathrm{prMD};L_\nu}$ obtaining the iterate $(x^{\nu+1}; t^{\nu+1})$. Iterate either for a finite number of iterations with increasing sample sizes (i.e., $L_{\nu+1} > L_\nu$), or until a prescribed termination criterion is satisfied (e.g., when L_ν is sufficiently large). □

A word of caution is in order about the practical termination of the algorithm. The basic goal of sample average approximation (SAA) schemes, on which the above algorithm is based, is to solve one single SAA subproblem corresponding to a (sufficiently large) sample size. The convergence theory guarantees that as the sample size tends to ∞, the estimators converge to the true solutions in an almost sure sense. Error analysis tells us how large the sample size should be for the error to be within a prescribed tolerance in a probabilistic sense. Thus SAA is not an algorithm in a conventional sense (no sequence generated). In the algorithm above, we generate a sequence of iterates corresponding to increasing sample sizes without a clean termination criterion. Generally, in schemes where we have almost sure convergence, we might terminate when the sample size is large (but we have no understanding of how large unless we derive error bounds).

We derive an almost-sure convergence result (Theorem 6) for the above algorithm by analyzing a convergent sequence of solutions $\{(x^L, t^L)\}$ that has a limit (x^∞, t^∞), where $x^L \triangleq \left(x^{L,i}\right)_{i=1}^N$, $t^L \triangleq \left(t_i^L\right)_{i=1}^N$ $x^\infty \triangleq \left(x^{\infty,i}\right)_{i=1}^N$, and $t^\infty \triangleq \left(t_i^\infty\right)_{i=1}^N$, and each pair (x^L, t^L) is a Nash equilibrium of the game $\mathcal{G}^{\mathrm{prMD};L}$. Similar to the analysis in [25, Section 5], a first step in claiming that (x^∞, t^∞) is a Nash equilibrium of the game $\mathcal{G}^{\mathrm{prMD}}$ is to establish the following limit for $\gamma \in \left\{\underline{\gamma}_i, \overline{\gamma}_i\right\}$:

$$\lim_{L\to\infty} \frac{1}{L} \sum_{\ell=1}^{L} H_{i;\lambda_i}(\psi_i(x^{L,i};\omega^\ell); t_i^L; \gamma)$$
$$= \mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x^{\infty,i};\widetilde{\omega}); t_i^\infty; \gamma) \right] \text{ almost surely;} \tag{33}$$

cf. [25, Part I, Theorem 6]. In turn, the proof of this equality (see also [71, Theorem 7.48]) is based on the following lemma.

Lemma 5 *Under assumptions (H1) and (H2) and the setting of Proposition 2, the following two statements hold.*

(a) *For all* $(x^i, t_i) \in X^i \times \mathcal{T}_i$*:*

$$\lim_{L\to\infty} \frac{1}{L} \sum_{\ell=1}^{L} H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell); t_i; \gamma) \;=\; \mathbb{E}\Big[\, H_{i;\lambda_i}(\psi_i(x^i;\widetilde{\omega}); t_i; \gamma)\,\Big] \text{ almost surely.} \tag{34}$$

(b) *A positive constant* c_i *exists such that for all* $\widehat{x}^i$ *and* $\widetilde{x}^i$ *in a compact subset of* X^i *and all* $\omega \in \Omega$ *for which* $\psi_i(\widehat{x}^i;\omega)$ *and* $\psi_i(\widetilde{x}^i;\omega)$ *are finite, and all* $\widehat{t}_i$ *and* $\widetilde{t}_i$ *in the compact interval* $\mathcal{T}_i$*:*

$$\Big|\, H_{i;\lambda_i}(\psi_i(\widehat{x}^i;\omega); \widehat{t}_i; \gamma) - H_{i,\lambda_i}(\psi_i(\widetilde{x}^i;\omega); \widetilde{t}_i; \gamma)\,\Big| \le c_i \,\Big[\|\widehat{x}^i - \widetilde{x}^i\| + |\widehat{t}_i - \widetilde{t}_i|\Big]. \tag{35}$$

Proof Similar to the proof of (24), it follows from assumption (H2) and part (c) of Proposition 2 that for all $\omega \in \Omega$,

$$\sup_{(x^i,t_i)\in X^i\times\mathcal{T}_i} |\, H_{i;\lambda_i}(\psi_i(x^i;\omega); t_i; \gamma)\,| \;\le\; \eta_{H_i}\,(\,1 + |\psi_i(x^i;\omega)|\,) \;\le\; \eta_{H_i}\,(\,1 + \widehat{\eta}_i\,).$$

This implies that the random variable $H_{i;\lambda_i}(\psi_i(x^i;\widetilde{\omega}); t_i; \gamma)$ has a finite variance that is bounded independently of $(x^i, t_i) \in X^i \times \mathcal{T}_i$. Since the family of random variables $\big\{H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell); t_i; \gamma)\big\}_{\ell=1}^{L}$ are independently and identically distributed samples of $H_{i;\lambda_i}(\psi_i(x^i;\widetilde{\omega}); t_i; \gamma)$, the limit (34) follows from the strong law of large numbers. Thus (a) holds. Part (b) follows easily from the Lipschitz continuity of $H_{i;\lambda_i}$ and part (e) of Proposition 2. □

We are now ready to formally state and prove the following convergence result of the above sequential sampling based algorithm.

Theorem 6 *Suppose that the set* $\mathbf{X}$ *is compact. Assume that each function* H_{i,λ_i} *satisfies conditions (H1) and (H2) and that the recourse functions* $\psi_i(x^i, \omega)$ *satisfies the assumptions of Proposition 2 for all* $\omega \in \Omega$*. Let* $\{L_\nu\}$ *be a sequence of positive integers tending to* ∞*. For each* ν*, let* $\{\omega^{\nu,1}, \ldots, \omega^{\nu,L_\nu}\}$ *be* L_ν *independently and identically distributed samples of the random variable* $\widetilde{\omega}$*; further let* (x^ν, t^ν) *be a NE of the sampled game* $\mathcal{G}^{\,\mathrm{prMD};L_\nu}$ *defined by these samples. If* (x^∞, t^∞) *is the limit of such a sequence of NE solutions, then* (x^∞, t^∞) *is almost surely a NE of the game* $\mathcal{G}^{\,\mathrm{prMD}}$.

Proof Similar to the proof of Theorem 6 in [25], we first show that (33) holds under (34) and (35). This in turn is not difficult in view of the following expression:

$$\begin{aligned}
&\frac{1}{L}\sum_{\ell=1}^{L} H_{i;\lambda_i}(\psi_i(x^{L,i};\omega^\ell);t_i^L;\gamma) - \mathbb{E}\Big[\, H_{i;\lambda_i}(\psi_i(x^{\infty,i};\widetilde{\omega});t_i^\infty;\gamma)\,\Big] \\
&= \left\{ \frac{1}{L}\sum_{\ell=1}^{L}\Big[\, H_{i;\lambda_i}(\psi_i(x^{L,i};\omega^\ell);t_i^L;\gamma) - H_{i;\lambda_i}(\psi_i(x^{\infty,i};\omega^\ell);t_i^\infty;\gamma)\,\Big]\right\} \\
&\quad + \left\{ \frac{1}{L}\sum_{\ell=1}^{L}\Big[\, H_{i;\lambda_i}(\psi_i(x^{\infty,i};\omega^\ell);t_i^\infty;\gamma)\Big] - \mathbb{E}\Big[\, H_{i;\lambda_i}(\psi_i(x^{\infty,i};\widetilde{\omega});t_i^\infty;\gamma)\,\Big]\right\}.
\end{aligned}$$

With (33) holding, the desired equilibrium property of the limit (x^∞, t^∞) follows from the minimizing property of each iterate (x^ν, t^ν), $\nu = 1, 2, \ldots$: for every $i = 1, \ldots, N$, and all $(x^i, t_i) \in X^i \times \mathcal{T}_i$,

$$\begin{aligned}
&\theta_i(x^i, x^{\nu-1,-i}) + \frac{1}{L_\nu}\max\Bigg(\sum_{\ell=1}^{L_\nu} H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell);t_i;\underline{\gamma}_i), \\
&\qquad \sum_{\ell=1}^{L_\nu} H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell);t_i;\overline{\gamma}_i)\Bigg) \\
&\geq \theta_i(x^{\nu;i}, x^{\nu-1;-i}) + \frac{1}{L_\nu}\max\Bigg(\sum_{\ell=1}^{L_\nu} H_{i;\lambda_i}(\psi_i(x^{\nu,i};\omega^\ell);t_i^\nu;\underline{\gamma}_i), \\
&\qquad \sum_{\ell=1}^{L_\nu} H_{i;\lambda_i}(\psi_i(x^{\nu,i};\omega^\ell);t_i^\nu;\overline{\gamma}_i)\Bigg).
\end{aligned}$$

Passing to the limit $\nu \to \infty$ readily completes the proof of the theorem. □

It is useful to note that unlike Theorem 4 which asserts the (deterministic) sequential convergence of the iterates produced by best response, Theorem 6 pertains to the (almost-sure) subsequential convergence of the iterates produced by the sampled games. It is not clear if sequential convergence could be established in the latter approach, even allowing convergence in an almost-sure sense.

5.2 A sampled best-response approach

One way to solve each sampled game $\mathcal{G}^{\mathrm{prMD};L_\nu}$ is through a best-response algorithm. Doing so will result in a double-loop algorithm where each outer loop corresponds to the selection of the samples $\{\omega^{\nu,\ell}\}_{\ell=1}^{L_\nu}$ and each inner loop consists of iteratively solving the sampled game defined by these samples. It is possible to derive a single-loop sampled algorithm that essentially implements a single best-response step in each inner loop. The resulting algorithm is a middle ground between the direct adaptation of the

deterministic best-response algorithm which requires solving stochastic optimization problems at each iteration and the sampled sub-game algorithm that solves a sequence of sampled games. In effect, this introduces a dependence between iterates that is absent in the previous algorithm. We formally state this sampled best-response algorithm below. In this algorithm, we write, for given samples $\{\omega^\ell\}_{\ell=1}^{L_\nu}$ in Ω,

$$\varphi_i^L(x^i;t_i) \triangleq \max\left(\frac{1}{L}\sum_{\ell=1}^{L}\left[\,H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell);t_i;\underline{\gamma}_i)\,\right],\right.$$
$$\left.\frac{1}{L_\nu}\sum_{\ell=1}^{L_\nu}\left[\,H_{i;\lambda_i}(\psi_i(x^i;\omega^\ell);t_i;\overline{\gamma}_i)\,\right]\right).$$

Sampled best-response algorithm for $\mathcal{G}^{\text{prMD}}$. At iteration $\nu = 0, 1, 2, \ldots$, given the samples $\{\omega^{\nu,\ell}\}_{\ell=1}^{L_\nu}$ in Ω and an iterate $x^\nu \triangleq \left(x^{\nu;i}\right)_{i=1}^N$ in $\mathbf{X}$ let each

$$(x^{\nu+1;i}, t_i^{\nu+1}) \in \operatorname*{argmin}_{x^i\in X^i,\, t_i\in\mathcal{T}_i}\left\{\theta_i\left(x^i, \underbrace{x^{\nu;-i}}_{\text{fixed in iteration}\,\nu}\right) + \varphi_i^{L_\nu}(x^i;t_i) + \tfrac{1}{2}\,\|\,x^i - x^{\nu;i}\,\|^2\right\} \tag{36}$$

for $i = 1, \ldots, N$. Iterate as per previous discussion. □

Neither Theorem 4 nor Theorem 6 is directly applicable to the above sampled best-response algorithm as the iterates $\{x^\nu, t_\nu\}$ are neither deterministic nor solutions to the sampled game $\mathcal{G}^{\text{prMD};L_\nu}$. Thus a separate analysis is needed. For one thing, such a proof would be complicated by the fact that the objective function in (36) is convex in the t_i-variable, albeit not strongly. Regularizing this variable in the subproblem (36) does not help the proof; instead it is necessary to regularize the sampled recourse function $\varphi_i^{L_\nu}(x^i;t_i)$ prior to setting up these optimization subproblems. Thus, rather than proving the convergence of this sampled best-response algorithm for the special case of private recourse functions at this time, we will postpone the proof until after we have introduced smoothing to induce differentiability into the recourse functions; see Sect. 6.6 for such a proof. We close by noting that an inexact best response scheme is also presented in our related work [68] where in each major iteration, all players take a single best-response step with a prescribed inexactness level. Across major iterations, we reduce the inexactness level, and under suitable conditions on the decay of the inexactness level, convergence statements are derived.

6 General case: recourse functions with coupling

Returning to the game $\mathcal{G}^{\text{rfMD}}$ where each recourse function $\psi_i(x;\omega)$ is defined by a convex QP (9) with the players' variables coupled in both the objective and constraints,

we discuss ways to deal with the challenges of this game so that eventually we can extend the algorithms for the case of private recourse to the general case. Ideally, we would like to address these challenges all at once and design a single-loop iterative scheme that solves a sequence of SPs or sampled games, each being an approximation of the original one wherein all the issues are approximated and updated in every iteration. However, one really cannot expect that such a single-loop scheme can be proved to be convergent. A more realistic way to deal with the challenges is to consider them individually, or perhaps by combining a couple, but not all, of them in an outer iterative scheme and then approximate the remaining one(s) in a sequence of inner iterations, each of which may require further sub-iterations. Even such a multi-loop scheme is not straightforward as there are different ways to carry out the inner versus the outer iterations. Extending the case of private recourse, we summarize the various approaches in this coupled case.

- Directly adopt a deterministic best-response scheme to the game $\mathcal{G}^{\text{rfMD}}$ wherein stochastic subprograms are solved; due to the aforementioned features of the players' optimization problem (26), such as non-smoothness of the objective functions and coupled variables in the recourse constraints, convergence of such a scheme cannot be easily proved as done in the case of private recourse. Indeed as required by the (DD) condition, the objective functions are expected to have mixed second derivatives. While this requirement can been relaxed somewhat (as presently being investigated in a separate work), once differentiability is still needed on the objective functions; this motivates the smoothing procedure described in Sect. 6.1.
- First smooth out all the non-smoothness in the players' objective functions and attempt to solve the resulting sampled games that still contain the expectation operation which needs to be discretized. At this juncture, there are 2 ways to deal with the resulting smoothed sampled games, each leading to a 3-loop scheme (with the smoothing being the outer-most scheme): (a) apply best-response iterations to the smoothed sampled games followed by sampling and approximation of the expectation operator; or (b) apply sampling and approximation of the expectation operator and then solve the discretized sampled games by existing game-theoretic methods, such as a best-response scheme.
- First apply a sampling and approximation procedure to the expectation operator, leading to a sequence of sampled games in which the expectation is discretized as finite sums derived from the sample approximation of the uncertainty. The solution of the resulting sampled games can then be attempted by existing game-theoretic methods, such as a best-response scheme which would require the sampled games to be smoothed. The overall approach requires essentially the same sampling/smoothing/sub-games operations as in the previous approach except that the sampling and smoothing steps are reversed.

A roadmap of the section. Aimed principally at the game $\mathcal{G}^{\text{rfMD}}$, the rest of this section is divided into six subsections. After formally introducing (Sect. 6.1) a logarithmic smoothing of the max function of two arguments followed by an illustration with the function $H_\lambda^{\text{QD}}(z; t; \gamma)$ and applying (Sect. 6.2) the regularization technique presented in Sect. 4.2 to each recourse function $\psi_i(x; \omega)$, we define the smoothed, regularized game $\mathcal{G}_s^{\text{srMD}}$ in Sect. 6.3. A proof of convergence of a Nash solution of the latter game to

a Nash solution of the original reformulated game $\mathcal{G}^{\text{rfMD}}$ as the (positive) parameter s (taken to be the same for both smoothing and regularization) tends to zero is established in Sect. 6.4. While the concern for non-smoothness and non-differentiability no longer exists in the smoothed, regularized game $\mathcal{G}_s^{\text{srMD}}$ for each fixed s, the expectation operator remains an issue that needs attention. This issue is addressed in the remaining two Sects. 6.5 and 6.6 that pertain to the solution of the game $\mathcal{G}_s^{\text{srMD}}$ for fixed s. Three approaches are sketched in the former subsection that paves the way into the latter subsection where sampling and best response schemes are jointly applied to the game $\mathcal{G}_s^{\text{srMD}}$; convergence is established for the joint sampling and best-response scheme in the case of private recourse function.

6.1 Smoothing

Keys to the remainder of the paper are the techniques of smoothing and regularization of the game $\mathcal{G}^{\text{rfMD}}$. For smoothing, we have the following functions to be smoothed: the max function (27) and its generalization to the abstract mean-deviation function $H_{i;\lambda_i}$, and the recourse function $\psi_i(x;\omega)$ via regularization. For regularization, in addition to the quadratic program (9) and/or its Lagrangian dual function (14), there is also the t_i-variable in the function $H_{i;\lambda_i}(z;t_i,\gamma)$. While we have not regularized the latter in the case of private course (see Sect. 5), it seems that this is needed in the present case.

We begin by introducing the smoothing of the 2-term pointwise maximum function (27) by a logarithmic function [18, Section 4]:

$$\max(a_1,a_2) \;\approx\; s\,\log\big[\,\exp(a_1/s)+\exp(a_2/s)\,\big],\; s\;>\;0.$$

Preserving the convexity of the max function, the approximation is an analytic function satisfying

$$\frac{\partial\left(s\,\log\big[\,\exp(a_1/s)+\exp(a_2/s)\,\big]\right)}{\partial a_i} \;=\; \frac{\exp(a_i/s)}{\exp(a_1/s)+\exp(a_2/s)} \in (\,0,1\,);\; i=1,2,$$

thus is Lipschitz continuous with a modulus upper bounded by unity. Moreover, the following inequalities can be easily derived (see [18, Proposition 5]): for all scalars a_1 and a_2,

$$0\;\le\; s\,\log\big[\,\exp(a_1/s)+\exp(a_2/s)\,\big]-\max(a_1,a_2)\;\le\; s\,\log 2,\; s\;>\;0. \tag{37}$$

Applying this approximation to the plus function $[a]_+=\max(a,0)$ results in a smooth function that preserves the non-increasing property of the plus function. To illustrate, consider the quantile-deviation function $H_\lambda^{\text{QD}}(z;t;\gamma)=t+(1-\lambda+\lambda\gamma)\,(z-t)+\lambda(z-t)_+$, which is approximated by the analytic function

$$\begin{aligned} H_{\lambda;s}^{\text{QD}}(z;t;\gamma) &= t+(\,1-\lambda+\lambda\,\gamma\,)\,(z-t)\\ &\quad +\lambda\,s\,\log\left[\,1+\exp\left(\frac{z-t}{s}\right)\right],\quad \text{for } s>0; \end{aligned} \tag{38}$$

the error satisfies: for all (z, t, γ),

$$0 \;\le\; H^{\mathrm{QD}}_{\lambda;s}(z;t;\gamma) - H^{\mathrm{QD}}_{\lambda}(z;t;\gamma) \;\le\; s\,\log 2,\; s \;>\; 0. \tag{39}$$

Moreover, it is not difficult to verify that the function $H^{\mathrm{QD}}_{\lambda;s}(z;t;\gamma)$ is convex in $(z;t)$ jointly for fixed γ and strictly convex in t for fixed $(z;\gamma)$. Notice that unlike the original $H^{\mathrm{QD}}_{\lambda}(z;t;\gamma)$ that is piecewise linear, thus not differentiable, in (z,t) for fixed γ, the approximation $H^{\mathrm{QD}}_{\lambda;s}(z;t;\gamma)$ is analytic in (z,t). This is part of the purpose of smoothing, namely, to remove non-differentiability.

In general, with $H_{i;\lambda_i}(z;t_i;\gamma_i)$ being an abstract mean-deviation function, we let the approximating function be denoted by $H_{i;\lambda_i;s}(z;t_i;\gamma_i)$, which is designed to smooth out the non-smoothness of the former function; in particular, we want $H_{i;\lambda_i;s}(z;t_i;\gamma_i)$ to be differentiable in (z,t_i) for fixed γ_i. Together with the above approximation of the 2-term pointwise max operator, we obtain the following approximation of (27):

$$\begin{aligned}\widehat{h}_{i;\lambda_i}(x;t_i) \approx s\,\log\Bigg\{&\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right]\right)\\ &+\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right)\Bigg\}\\ \triangleq\; &\widehat{h}_{i,\lambda_i;s}(x;t_i),\end{aligned}$$

where $H_{i;\lambda_i;s}(z;t_i,\gamma_i) \approx H_{i;\lambda_i}(z;t_i;\gamma_i)$ and $\widehat{h}_{i,\lambda_i;s}(x;t_i) \triangleq \widehat{H}_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i)$ with

$$\begin{aligned}\widehat{H}_{i;\lambda_i;s}(\mathcal{Z};t_i) \;\triangleq\; s\,\log\Bigg\{&\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\mathcal{Z};t_i;\underline{\gamma}_i)\right]\right)\\ &+\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\mathcal{Z};t_i;\overline{\gamma}_i)\right]\right)\Bigg\}.\end{aligned}$$

Overall, the approximating function $H_{i;\lambda_i;s}(z;t_i,\gamma_i)$. We summarize properties of this approximation function as follows:

Assumption ($\mathbf{H}^{\mathrm{apprx}}$): $H_{i;\lambda_i;s}(z;t_i,\gamma_i)$ is continuously differentiable and convex in (z,t_i) for fixed γ_i; strictly increasing in z for fixed (t,γ_i); and linear in γ_i for fixed (z,t); finally, there exist scalars $\delta_i > 0$ and $\overline{s}_i > 0$ such that $|H_{i;\lambda_i;s}(z;t_i;\gamma_i) - H_{i;\lambda_i}(z;t_i;\gamma_i)| \le s\,\delta_i$ for all such $(s;z;t_i;\gamma_i)$; the derivative $\left|\dfrac{\partial H_{i;\lambda_i;s}(z;t_i,\gamma_i)}{\partial z}\right|$ is upper bounded by $\nabla H_i^{\mathrm{ub}} > 0$ for all $s \in (0,\overline{s}_i)$, all $(t_i;\gamma_i) \in \mathcal{T}_i \times \Gamma_i$, and z of interest. □

We illustrate this approximation with the mean-quantile deviation function $H^{\mathrm{QD}}_{\lambda}(z;t;\gamma)$ that is approximated by the function $H^{\mathrm{QD}}_{\lambda;s}(z;t;\gamma)$ given by (38). Specifically, we verify that the latter function satisfies Assumption ($\mathrm{H}^{\mathrm{apprx}}$) provided that both λ and γ are in the interval $(0,1)$. Clearly this function is continuously differentiable. To see that it is convex in (z,t) for fixed γ, it suffices to verify that the bivariate

function $f(u,v) \triangleq \log\left[1+\exp\left(\frac{u-v}{s}\right)\right]$ is convex on $\mathbb{R}^2$ for fixed scalar $s>0$. We have

$$\frac{\partial f(u,v)}{\partial u} = \frac{1}{s}\left[1-\frac{1}{1+\exp\left(\frac{u-v}{s}\right)}\right] \text{ and } \frac{\partial f(u,v)}{\partial v} = -\frac{1}{s}\left[1-\frac{1}{1+\exp\left(\frac{u-v}{s}\right)}\right].$$

Hence

$$\frac{\partial^2 f(u,v)}{\partial u^2} = \frac{1}{s^2}\frac{\exp\left(\frac{u-v}{s}\right)}{\left\{1+\exp\left(\frac{u-v}{s}\right)\right\}^2} = \frac{\partial^2 f(u,v)}{\partial v^2} = -\frac{\partial^2 f(u,v)}{\partial u\,\partial v}.$$

The latter expressions are enough to establish the convexity of $f(u,v)$. The bound (39) easily yields $|H^{\mathrm{QD}}_{\lambda;s}(z;t;\gamma) - H^{\mathrm{QD}}_{\lambda}(z;t;\gamma_i)| \le s\,\log 2$. Finally,

$$\begin{aligned} 1+\lambda\gamma > \frac{\partial H^{\mathrm{QD}}_{\lambda;s}(z;t,\gamma)}{\partial z} &= (1-\lambda+\lambda\gamma) + \lambda\left[1-\frac{1}{1+\exp\left(\frac{z-t}{s}\right)}\right] \\ &\ge 1-\lambda+\lambda\gamma > 0 \end{aligned}$$

In general, with $H_{i;\lambda_i}$ satisfying assumption (H2), so does $H_{i;\lambda_i}$. Indeed we have, for $\widehat{\eta}_i \triangleq \max(\eta_i, \overline{s}_i\,\delta)$

$$\begin{aligned} |\,H_{i;\lambda_i;s}(z;t_i;\gamma_i)\,| &\le |\,H_{i;\lambda_i}(z;t_i;\gamma_i)\,| + |\,H_{i;\lambda_i;s}(z;t_i;\gamma_i) - H_{i;\lambda_i}(z;t_i;\gamma_i)\,| \\ &\le \eta_i\,(1+\|\,z\,\|) + s\,\delta_i \;\le\; \widehat{\eta}_i\,(1+\|\,z\,\|), \end{aligned}$$

for all $s \in (0,\overline{s}_i)$, all $(t_i,\gamma_i) \in \mathcal{T}_i \times \Gamma_i$ and all z of interest. Moreover, by (37), we have

$$\begin{aligned} 0 \le s\,\log\left\{\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\mathcal{Z};t_i;\underline{\gamma}_i)\right]\right) + \exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\mathcal{Z};t_i;\overline{\gamma}_i)\right]\right)\right\} \\ - \max\left\{\mathbb{E}\left[\,H_{i;\lambda_i;s}(\mathcal{Z};t_i;\underline{\gamma}_i)\right],\ \mathbb{E}\left[\,H_{i;\lambda_i;s}(\mathcal{Z};t_i;\overline{\gamma}_i)\right]\right\} \;\le\; s\,\log 2. \end{aligned}$$

Since

$$
\begin{aligned}
&\widehat{h}_{i,\lambda_i;s}(x;t_i) - \widehat{h}_{i;\lambda_i}(x;t_i) \\
&\quad = s\,\log\Bigg\{\exp\Bigg(\mathbb{E}\Bigg[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Bigg]\Bigg) \\
&\qquad + \exp\Bigg(\mathbb{E}\Bigg[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\Bigg]\Bigg)\Bigg\} \\
&\qquad - \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\} \\
&\quad = s\,\log\Bigg\{\exp\Bigg(\mathbb{E}\Bigg[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Bigg]\Bigg) \\
&\qquad + \exp\Bigg(\mathbb{E}\Bigg[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\Bigg]\Bigg)\Bigg\} \\
&\qquad - \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\} \\
&\qquad + \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\} \\
&\qquad - \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\},
\end{aligned}
$$

which yields,

$$
\begin{aligned}
&\max\Big\{\mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\} \\
&\quad - \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\} \\
&\le \widehat{h}_{i,\lambda_i;s}(x;t_i) - \widehat{h}_{i;\lambda_i}(x;t_i) \\
&\le s\,\log 2 + \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\} \\
&\quad - \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\},
\end{aligned}
$$

applying the elementary inequality

$$
\begin{aligned}
|\max(a,b) - \max(c,d)| &\le |\max(a-b,0) - \max(c-d,0)| + |b-d| \\
&\le 2\,(|a-c| + |b-d|)
\end{aligned}
\tag{40}
$$

to the difference

$$
\begin{aligned}
&\Big|\max\Big\{\mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i;s}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\} \\
&\quad - \max\Big\{\mathbb{E}\Big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big],\ \mathbb{E}\big[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\big]\Big\}\Big|.
\end{aligned}
$$

We deduce from assumption (H^{apprx}) that for all $x \in \mathbf{X}$ and all $s \in (0, \overline{s}_i)$,

$$
|\widehat{h}_{i,\lambda_i;s}(x;t_i) - \widehat{h}_{i;\lambda_i}(x;t_i)| \le s\,(4\,\delta_i + \log 2)\,. \tag{41}
$$

This bound will be extended in the next subsection when we regularize the recourse function $\psi_i(x;\omega)$.

6.2 Regularization

Using the approximating function $\widehat{h}_{i;\lambda_i;s}(x;t_i)$ which still contains the non-differentiable recourse functions $\psi_i(x;\omega)$ implicitly, player i's optimization problem is

$$\underset{x^i\in X^i,\, t_i\in\mathcal{T}_i}{\text{minimize}} \left(\theta_i(x^i,x^{-i}) + \widehat{h}_{i;\lambda_i;s}(x;t_i)\right). \tag{42}$$

While being a convex program in the (x^i,t_i)-variables, the non-differentiability of the function $\widehat{h}_{i;\lambda_i;s}(x;t_i)$ complicates the convergence proof of a best-response scheme for solving the approximate game. As a first step, we assume that the family of functions $\{\theta_i(x)\}_{i=1}^N$ satisfies assumption (DD). Ideally, we want this assumption to hold for the sum objective in (42); yet the function $\widehat{h}_{i;\lambda_i;s}(x;t_i)$ is not even once differentiable, let alone twice differentiable in the pair of variables x^i and t_i (and in x^j for $j\neq i$), as required by the condition (DD). Thus, we are led to the regularization of the convex QP (9) that defines the recourse function $\psi_i(x;\omega)$. Following the development in Sect. 4.2, we consider the following regularized dual program: for $s>0$

$$\begin{aligned}
&\psi_{i;s}(x;\omega)\\
&\quad\triangleq \underset{z^i,\,v^i}{\text{maximum}} \left[\xi^i(\omega) - \sum_{j=1}^N C^{ij}(\omega)x^j\right]^T v^i - \tfrac{1}{2}(z^i)^T Q^i z^i\\
&\qquad -\frac{s}{2}\left[(z^i)^T z^i + (v^i)^T v^i\right]\\
&\quad\text{subject to}\quad -Q^i z^i + (D^i)^T v^i = f^i(\omega) + \sum_{j\neq i} G^{ij}(\omega)x^j \text{ and } v^i \geq 0.
\end{aligned} \tag{43}$$

6.3 Smoothed and regularized game $\mathcal{G}_s^{\text{srMD}}$

After applying smoothing and regularization to the reformulated game $\mathcal{G}^{\text{rfMD}}$, we obtain the smoothed and regularized game $\mathcal{G}_s^{\text{srMD}}$ that is composed of the optimization problems:

$$\left\{\underset{x^i\in X^i,\, t_i\in\mathcal{T}_i}{\text{minimize}} \left(\underbrace{\theta_i(x^i,x^{-i}) + \widehat{h}^{\text{apprx}}_{i;\lambda_i;s}(x;t_i) + \frac{s}{2}t_i^2}_{\widehat{\theta}_{i;s}(x;t_i)}\right)\right\}_{i=1}^N, \tag{44}$$

where $\widehat{h}^{\text{apprx}}_{i;\lambda_i;s}(x;t_i) \triangleq \widehat{H}_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i)$ is given by

$$s\,\log\left\{\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right]\right)\right.$$
$$\left.+\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right)\right\},$$

which is finite, continuous, and differentiable in x^i for almost all $\omega \in \Omega$, under the assumptions of Proposition 2. The result below summarizes properties of the approximating function $\widehat{h}^{\text{apprx}}_{i;\lambda_i;s}(x;t_i)$.

Proposition 7 *Let Q^i be a symmetric positive semidefinite matrix for every i. Suppose that the assumptions of Proposition 2 and assumptions (H2) and (H^{apprx}) hold. Then the functions $\widehat{h}_{i;\lambda_i}(x;t_i)$ and $\widehat{h}^{\text{apprx}}_{i;\lambda_i;s}(x;t_i)$ are continuous on $\mathbf{X}\times\mathcal{T}_i$ for every i; moreover, there exists a constant $\sigma_i > 0$ such that for all $s \in (0,\bar{s}_i)$ and all $(x,t_i) \in \mathbf{X}\times\mathcal{T}_i$,*

$$\left|\widehat{h}^{\text{apprx}}_{i;\lambda_i;s}(x;t_i) - \widehat{h}_{i;\lambda_i}(x;t_i)\right| \le s\,\sigma_i\left[1+\|x\|+\|x\|^2\right]. \tag{45}$$

Lastly, $\nabla_{x^i}\widehat{h}^{\text{apprx}}_{i;\lambda_i;s}(x^i,x^{-i};t_i)$ exists and is continuous.

Proof By part (c) of Proposition 3, we have, for some constant $c_i > 0$, dependent on the pair (Q^i, D^i) only, such that for all $(x;\omega)$ for which $\psi_i(x;\omega)$ is finite,

$$|\,\psi_{i;s}(x;\omega) - \psi_i(x;\omega)\,| \le s\,c_i\left[\left\|\xi^i(\omega) - \sum_{j=1}^{N} C^{ij}(\omega)x^j\right\|\right.$$
$$\left.+\left\|f^i(\omega) + \sum_{j\neq i} G^{ij}(\omega)x^j\right\|\right]^2,$$

Similar to the derivation of (41), we have

$$\begin{aligned}
&\widehat{h}^{\text{apprx}}_{i,\lambda_i;s}(x;t_i) - \widehat{h}_{i;\lambda_i}(x;t_i)\\
&\quad= s\,\log\left\{\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right]\right)\right.\\
&\qquad\left.+\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right)\right\}\\
&\qquad-\max\left\{\mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right],\ \mathbb{E}\left[H_{i;\lambda_i}(\psi_i(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right\}\\
&\quad= s\,\log\left\{\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right]\right)\right.\\
&\qquad\left.+\exp\left(\mathbb{E}\left[\frac{1}{s}\,H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right)\right\}
\end{aligned}$$

$$
\begin{aligned}
&- \max\left\{ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right],\ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right\}\\
&+ \max\left\{ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right],\ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right\}\\
&- \max\left\{ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right],\ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right\}\\
&+ \max\left\{ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right],\ \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right\}\\
&- \max\left\{ \mathbb{E}\left[H_{i;\lambda_i}(\psi_{i}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\right],\ \mathbb{E}\left[H_{i;\lambda_i}(\psi_{i}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right\}.
\end{aligned}
$$

Under assumptions ($\mathrm{H}^{\mathrm{apprx}}$) and (16) and (17) in Proposition 2, that for all $x \in \mathbf{X}$ and all $t_i \in \mathcal{T}_i$, we deduce, using (40) again,

$$
\begin{aligned}
&|\,\widehat{h}^{\mathrm{apprx}}_{i;\lambda_i;s}(x;t_i) - \widehat{h}_{i;\lambda_i}(x;t_i)\,|\\
&\quad\le s\,(4\,\delta_i + \log 2) + 2\,\Big|\,\mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big]\\
&\qquad - \mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_{i}(x;\widetilde{\omega});t_i;\underline{\gamma}_i)\Big]\Big|\\
&\qquad + 2\,\Big|\,\mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right] - \mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i}(x;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\Big|\,.
\end{aligned}
$$

Hence (45) holds for an appropriate constant σ_i. Lastly, to show the continuous differentiability of $\widehat{h}^{\mathrm{apprx}}_{i;\lambda_i;s}(x^i, x^{-i};t_i)$ in x^i, it suffices to show the same for the expected composite function:

$$
\mathbb{E}\Big[H_{i;\lambda_i;s}(\psi_{i;s}(x^i, x^{-i};\widetilde{\omega});t_i;\gamma_i)\Big]
$$

for $\gamma_i \in \{\underline{\gamma}_i, \overline{\gamma}_i\}$. For this purpose, we note that for almost all ω and all $x \in \mathbf{X}$,

$$
\begin{aligned}
&\nabla_{x_i} H_{i;\lambda_i;s}(\psi_{i;s}(x^i, x^{-i};\omega);t_i;\gamma_i)\\
&\quad = \frac{\partial H_{i;\lambda_i;s}(z;t_i;\gamma_i)}{\partial z}\,|_{z=\psi_{i;s}(x;\omega)}\,\nabla_{x^i}\psi_{i;s}(x^i, x^{-i};\omega)
\end{aligned}
$$

By assumption ($\mathrm{H}^{\mathrm{apprx}}$), we have $\left|\dfrac{\partial H_{i;\lambda_i;s}(z;t_i;\gamma_i)}{\partial z}\,|_{z=\psi_{i;s}(x;\omega)}\right| \le \nabla H_i^{\mathrm{ub}}$. With $\widehat{v}^{i;s}$ denoting the unique optimal solution of (43), we have $\nabla_{x^i}\psi_{i;s}(x^i, x^{-i};\omega) = -C^{ii}(\omega)^T\widehat{v}^{i;s}$ by Danskin's Theorem. Hence by (12), for some positive constant $c_i^{Q_i,D_i;s}$, for all $x \in \mathbf{X}$, and almost all $\omega \in \Omega$,

$$
\begin{aligned}
\Big\|\,\nabla_{x^i}\psi_{i;s}(x^i, x^{-i};\omega)\,\Big\| \le \Big(\operatorname{essup}|C^{ii}(\widetilde{\omega})|\Big)\,c_i^{Q_i,D_i;s}\,\Bigg[\,&\Big\|\,\xi^i(\omega) - \sum_{j=1}^{N} C^{ij}(\omega)x^j\,\Big\|\\
&+ \Big\|\,f^i(\omega) + \sum_{j\neq i} G^{ij}(\omega)x^j\,\Big\|\,\Bigg].
\end{aligned}
$$

This is enough to justify the interchange of differentiation with expectation, yielding

$$\begin{aligned}&\nabla_{x^i}\left\{\mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x^i,x^{-i};\widetilde{\omega});t_i;\gamma_i)\right]\right\}\\&=\mathbb{E}\left[\frac{\partial H_{i;\lambda_i;s}(z;t_i;\gamma_i)}{\partial z}\,|_{z=\psi_{i;s}(x;\widetilde{\omega})}\,\nabla_{x^i}\psi_{i;s}(x^i,x^{-i};\widetilde{\omega})\right].\end{aligned}$$

□

The bound (45) provides the basis for showing the convergence of an approximate Nash solution of the game $\mathcal{G}_s^{\mathrm{srMD}}$ to a Nash solution of the game $\mathcal{G}^{\mathrm{rfMD}}$ as the smoothing parameter $s \downarrow 0$.

6.4 The convergence of $\mathcal{G}_s^{\mathrm{srMD}}$ to $\mathcal{G}^{\mathrm{rfMD}}$

The following theorem establishes the stated convergence of a sequence of smoothed games to the reformulated game.

Theorem 8 *Under the setting of subsection 6.3, let the assumptions of Proposition 7 hold. For an arbitrary sequence of positive scalars $\{s_k\}$ converging to zero, let $\left\{(x^*_{s_k},t^*_{s_k})\right\}$ be a corresponding sequence such that each $(x^*_{s_k},t^*_{s_k})$ is a Nash solution of the game $\mathcal{G}_{s_k}^{\mathrm{srMD}}$ for every k. If $\left\{(x^*_{s_k},t^*_{s_k})\right\}$ converges to a limit (x^∞,t^∞), then (x^∞,t^∞) is a Nash solution of the original game $\mathcal{G}^{\mathrm{rfMD}}$.*

Proof The proof of this claim is not hard provided that we can demonstrate the uniform convergence of the family of approximants $\left\{\left\{\widehat{h}^{\mathrm{apprx}}_{i;\lambda;s_k}(x;t_i)\right\}_{k=1}^{\infty}\right\}_{i=1}^{N}$ to the family of players' mean-deviation functions $\left\{\widehat{h}_{i;\lambda_i}(x,t_i)\right\}_{i=1}^{N}$; see [25,30,31]. By definition, this entails showing the following condition for every $i=1,\ldots,N$:

- for every sequence $\{(x^{k;i};t_i^k)\}\subset X^i\times\mathcal{T}_i$ converging to $(x^{\infty;i};t_i^\infty)$

$$\lim_{k\to\infty}\widehat{h}^{\mathrm{apprx}}_{i;\lambda_i;s_k}(x^k;t_i^k)=\widehat{h}_{i;\lambda_i}(x^\infty;t_i^\infty),\ \text{where } x^k\triangleq(x^{k;j})_{j=1}^N. \tag{46}$$

The desired limit is easy to verify provided that (45) is in place. Indeed, we have

$$\begin{aligned}\widehat{h}^{\mathrm{apprx}}_{i;\lambda_i;s_k}(x^k;t_i^k)-\widehat{h}_{i;\lambda_i}(x^\infty;t_i^\infty)&=\left[\widehat{h}^{\mathrm{apprx}}_{i;\lambda_i;s_k}(x^k;t_i^k)-\widehat{h}_{i;\lambda_i}(x^k;t_i^k)\right]\\&\quad+\left[\widehat{h}_{i;\lambda_i}(x^k;t_i^k)-\widehat{h}_{i;\lambda_i}(x^\infty;t_i^\infty)\right].\end{aligned}$$

The inequality (45) implies that

$$\lim_{k\to\infty}\left[\widehat{h}^{\mathrm{apprx}}_{i;\lambda_i;s_k}(x^k;t_i^k)-\widehat{h}_{i;\lambda_i}(x^k;t_i^k)\right]=0,$$

and the continuity of $\widehat{h}_{i;\lambda_i}$ implies that

$$\lim_{k\to\infty}\left[\widehat{h}_{i;\lambda_i}(x^k;t_i^k)-\widehat{h}_{i;\lambda_i}(x^\infty;t_i^\infty)\right]=0.$$

The combination of the last two limits establishes (46). □

Theorem 8 specifies that if the sequence is convergent then its limit point is a NE of the original game. This theorem has left open the question of whether the sequence is indeed convergent. Further research is needed to address this question.

At this point, we have prescribed the outer loop in solving the game $\mathcal{G}^{\text{rfMD}}$; namely, we solve a sequence of smoothed and regularized sub-games $\left\{\mathcal{G}_{s_k}^{\text{srMD}}\right\}$ for a sequence of positive scalars $\{s_k\}$ tending to zero. The next step pertains to the solution of each such sub-game.

6.5 Solving the smoothed game $\mathcal{G}_s^{\text{srMD}}$ for fixed s

As in the case of private recourse, there are 3 iterative approaches for solving the game $\mathcal{G}_s^{\text{srMD}}$ for a fixed s. First is a direct application of the deterministic best-response algorithm that solves SPs at each iteration. Due to the once, but not twice, differentiability of each objective function $\widehat{\theta}_{i;s}(x;t_i)$ in (44) it is not possible to directly apply condition (DD) to infer the convergence of this direct scheme. An ongoing research aims at extending this condition to bypasses the requirement of twice differentiability. Nevertheless, this extension is beyond the scope of this paper.

The next approach to solve $\mathcal{G}_s^{\text{srMD}}$ for a fixed s is to apply the sequential sampling algorithm wherein a sequence of sampled games $\mathcal{G}_s^{\text{srMD};L}$ for $L=1,2,\ldots$, each corresponding to a set of samples in the current iteration, is solved by a deterministic method. Each $\mathcal{G}_s^{\text{srMD};L}$ is similar to $\mathcal{G}^{\text{prMD};L}$ and consists of the players' smoothed optimization problems given by:

$$\begin{aligned}&\Bigg\{\underset{(x^i,t_i)\in X^i\times\mathcal{T}_i}{\text{minimize}}\Bigg[\theta_i(x^i,x^{-i})+\frac{s}{2}t_i^2\\&\quad+s\,\log\Bigg\{\exp\Bigg(\frac{1}{L}\Bigg[\frac{1}{s}\sum_{\ell=1}^{L}H_{i;\lambda_i;s}(\psi_{i;s}(x;\omega^\ell);t_i;\underline{\gamma}_i)\Bigg]\Bigg)\\&\quad+\exp\Bigg(\frac{1}{L}\Bigg[\frac{1}{s}\sum_{\ell=1}^{L}H_{i;\lambda_i;s}(\psi_{i;s}(x;\omega^\ell);t_i;\overline{\gamma}_i)\Bigg]\Bigg)\Bigg\}\Bigg\}_{i=1}^{N}\end{aligned}$$

To prove the extension of Theorem 6 to this case of coupled recourse functions, it suffices to establish the extension of (33) to the following limit:

$$\begin{aligned}&\lim_{L\to\infty}\frac{1}{L}\sum_{\ell=1}^{L}H_{i;\lambda_i;s}(\psi_{i;s}(x^L;\omega^\ell);t_i^L;\gamma)\\&\quad=\mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x^\infty;\widetilde{\omega});t_i^\infty;\gamma)\right]\text{ almost surely.}\end{aligned}$$

In view of the Lipschitz continuity of the function $\psi_{i;s}(\bullet;\omega)$ asserted by part (e) of Proposition 2, the proof of the above limit is the same as before and is not repeated.

The benefits of smoothing and regularization lie in the solution of sampled game $\mathcal{G}_s^{\mathrm{srMD};L}$ in which the players' objective functions are once continuously differentiable. Thus, like the previous direct approach, one can now apply a best-response algorithm to $\mathcal{G}_s^{\mathrm{srMD};L}$; nevertheless, the same convergence challenge remains, namely, the lack of differentiability of the $\nabla_{x^i}\psi_i(\bullet;\omega)$ that has all the players' variables coupled. Further work is needed to complete the analysis of this solution approach.

In summary, we observe that the two solution approaches for solving the game $\mathcal{G}_s^{\mathrm{srMD}}$ for fixed s outlined above (a direct best-response approach and a sampled approach followed by best response) both suffer from the lack of twice differentiability of the players' objective functions when there is coupling of variables in the players' recourse functions, making the analysis of a fully convergent scheme for solving this game incomplete. In the next section, we complete this task for the same game by assuming a private recourse for each player.

6.6 Sampled best response for smoothed private recourse

So far the solution of the game $\mathcal{G}^{\mathrm{rfMD}}$ consists of two loops: the outer loop is the iterative construction of the smoothed games $\mathcal{G}_{s_k}^{\mathrm{srMD}}$ for a sequence of positive scalars $\{s_k\}$ tending to zero. The inner loop is the actual solution of each smooth game $\mathcal{G}_s^{\mathrm{srMD}}$ for fixed scalar s by some iterative scheme; the latter game remains a SNEP of the same type except that the players' mean-risk composite functions have been smoothed and their overall objectives have been strongly convexified, including the auxiliary variables t_i see (44). In the previous subsection, we have described a sampling method for solving the latter game with fixed s, resulting in an overall 3-loop scheme for solving $\mathcal{G}^{\mathrm{rfMD}}$ as described in the beginning of this section. In what follows, we revisit the iterative sampled best-response algorithm and apply it to the special case of the smoothed game $\mathcal{G}_s^{\mathrm{srMD}}$ with private resource for fixed s, this results in an overall 2-loop scheme for solving this special case of the un-smoothed game $\mathcal{G}^{\mathrm{rfMD}}$. The difference between the discussion here and that in Sect. 5.2 is that the algorithm there is a single-loop scheme for solving the same special case of the un-smoothed game $\mathcal{G}^{\mathrm{rfMD}}$ but no convergence proof was provided; here a double-loop scheme is proposed with convergence. In short, to obtain a convergent sampled best-response algorithm to solve the game $\mathcal{G}^{\mathrm{rfMD}}$, it is necessary to iterate in two steps: an outer step of smoothing and strongly convexification of this game and an inner step of solving a sampled stochastic program. For ease of reference in the convergence proof, we state the sampled best-response scheme applied to the smoothed game $\mathcal{G}_s^{\mathrm{smMD}}$ for a fixed s. Extending the notation $\varphi_i^L(x^i, t_i)$ used in Sect. 5.2 prior to smoothing, we let

$$\varphi_{i;s}^{L}(x^i;t_i \triangleq s\,\log\left\{\exp\left(\frac{1}{L}\left[\frac{1}{s}\sum_{\ell=1}^{L}H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\omega^\ell);t_i;\underline{\gamma}_i)\right]\right)\right.$$
$$\left.+\exp\left(\frac{1}{L}\left[\frac{1}{s}\sum_{\ell=1}^{L}H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\omega^\ell);t_i;\overline{\gamma}_i)\right]\right)\right\}$$

which is a random function that depends on the samples $\{\omega^\ell\}_{\ell=1}^{L}$ on hand.

Sampled best-response algorithm for $\mathcal{G}_s^{\mathrm{srMD}}$ with fixed s. At iteration $\nu = 0, 1, 2, \ldots$, given the sample size L_ν and samples $\{\omega^{\nu,\ell}\}_{\ell=1}^{L_\nu}$ in Ω and an iterate $x^\nu \triangleq \left(x^{\nu;i}\right)_{i=1}^{N}$ in $\mathbf{X}$ let each

$$(x^{\nu+1,i}, t_{\nu+1}) \in \underset{x^i\in X^i,\, t_i\in\mathcal{T}_i}{\operatorname{argmin}}\left\{\theta_i(x^i, x^{\nu;-i}) + \frac{s}{2}t_i^2 + \underbrace{\varphi_{i;s}^{L_\nu}(x^i;t_i)}_{\text{sample dependent}} + \tfrac{1}{2}\|x^i - x^{\nu;i}\|^2\right\}, \tag{47}$$

for $i = 1, \ldots, N$. □

Note that both smoothing and regularization have been applied to the optimization in (47). Before proving the convergence of the above algorithm in Theorem 10, we present a supporting lemma that is a consequence of the uniform law of large numbers [71, Theorem 7.48].

Lemma 9 *Under the assumptions of Theorem 10, it holds that*

$$\mathbb{P}\left\{\lim_{L\to\infty}\left[\sup_{(x^i,t_i)\in X^i\times\mathcal{T}_i}\left\|\begin{pmatrix}\nabla_{x^i}\varphi_{i;s}^{L}(x^i;t_i) - \nabla_{x^i}\varphi_{i;s}(x^i;t_i)\\ \dfrac{\partial\varphi_{i;s}^{L}(x^i;t_i)}{\partial t_i} - \dfrac{\partial\varphi_{i;s}(x^i;t_i)}{\partial t_i}\end{pmatrix}\right\|\right] = 0\right\} = 1, \tag{48}$$

where

$$\varphi_{i;s}(x^i;t_i) \triangleq s\,\log\left\{\exp\left\{\frac{1}{s}\mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\widetilde{\omega});t_i;\underline{\gamma}_i)\right]\right\}\right.$$
$$\left.+\exp\left\{\frac{1}{s}\mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\widetilde{\omega});t_i;\overline{\gamma}_i)\right]\right\}\right\}.$$

Proof To simplify the notation somewhat, we write

$$\underline{A}^L_{i;s}(x^i, t_i) \triangleq \exp\left\{\frac{1}{s}\frac{1}{L}\left[\sum_{\ell=1}^{L} H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\omega^\ell); t_i; \underline{\gamma}_i)\right]\right\}$$
$$\overline{A}^L_{i;s}(x^i, t_i) \triangleq \exp\left\{\frac{1}{s}\frac{1}{L}\left[\sum_{\ell=1}^{L} H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\omega^\ell); t_i; \overline{\gamma}_i)\right]\right\},$$

which are defined with respect to given samples $\{\omega^\ell\}_{\ell=1}^L$. We then have

$$\varphi^L_{i;s}(x^i; t_i) \;=\; s\,\log\left\{\underline{A}^L_{i;s}(x^i; t_i) + \overline{A}^L_{i;s}(x^i; t_i)\right\}.$$

Similarly, let

$$\underline{A}_{i;s}(x^i, t_i) \triangleq \exp\left\{\mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\widetilde{\omega}); t_i; \underline{\gamma}_i)\right]\right\}$$
$$\overline{A}_{i;s}(x^i, t_i) \triangleq \exp\left\{\mathbb{E}\left[H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\widetilde{\omega}); t_i; \overline{\gamma}_i)\right]\right\}$$

so that $\varphi_{i;s}(x^i; t_i) = s\,\log\left\{\underline{A}_{i;s}(x^i; t_i) + \overline{A}_{i;s}(x^i; t_i)\right\}$. By the chain rule of differentiation, we have

$$\nabla_{x^i}\varphi^L_{i;s}(x^i; t_i) \;=\; \frac{1}{L}\sum_{\ell=1}^{L}\nabla_{x^i}\psi_{i;s}(x^i;\omega^\ell)$$
$$\times\left[\underbrace{\frac{\dfrac{\partial H_{i;\lambda_i;s}(z; t_i; \underline{\gamma}_i)}{\partial z}\Big|_{z=\psi_{i;s}(x^i;\omega^\ell)}\underline{A}^L_{i;s}(x^i; t_i) + \dfrac{\partial H_{i;\lambda_i;s}(z; t_i; \overline{\gamma}_i)}{\partial z}\Big|_{z=\psi_{i;s}(x^i;\omega^\ell)}\overline{A}^L_{i;s}(x^i; t_i)}{\underline{A}^L_{i;s}(x^i; t_i) + \overline{A}^L_{i;s}(x^i; t_i)}}_{\text{sampled scalar}}\right]$$
$$\nabla_{x^i}\varphi_{i;s}(x^i; t_i) = \mathbb{E}\Bigg[\nabla_{x^i}\psi_{i;s}(x^i;\widetilde{\omega})$$
$$\times\frac{\dfrac{\partial H_{i;\lambda_i;s}(z; t_i; \underline{\gamma}_i)}{\partial z}\Big|_{z=\psi_{i;s}(x^i;\widetilde{\omega})}\underline{A}_{i;s}(x^i; t_i) + \dfrac{\partial H_{i;\lambda_i;s}(z; t_i; \overline{\gamma}_i)}{\partial z}\Big|_{z=\psi_{i;s}(x^i;\widetilde{\omega})}\overline{A}_{i;s}(x^i; t_i)}{\underline{A}_{i;s}(x^i; t_i) + \overline{A}_{i;s}(x^i; t_i)}\Bigg].$$

Thus we see that $\nabla_{x^i}\varphi^L_{i;s}(x^i; t_i)$ is a sample average approximation of $\nabla_{x^i}\varphi_{i;s}(x^i; t_i)$ with the samples $\{\omega^\ell\}_{\ell=1}^L$ on hand. According to [71, Theorem 7.48],

$$\mathbb{P}\left\{\lim_{L\to\infty}\left[\sup_{(x^i,t_i)\in X^i\times\mathcal{T}_i}\left\|\nabla_{x^i}\varphi^L_{i;s}(x^i; t_i) - \nabla_{x^i}\varphi_{i;s}(x^i; t_i)\right\|\right] = 0\right\} = 1,$$

holds if (a) the (random) function

$$
\begin{aligned}
&F(x^i;t_i;\omega)\\
&\quad\triangleq \nabla_{x^i}\psi_{i;s}(x^i;\omega)\\
&\qquad\times\frac{\dfrac{\partial H_{i;\lambda_i;s}(z;t_i;\underline{\gamma}_i)}{\partial z}\Big|_{z=\psi_{i;s}(x^i;\omega)}\underline{A}_{i;s}(x^i;t_i)+\dfrac{\partial H_{i;\lambda_i;s}(z;t_i;\overline{\gamma}_i)}{\partial z}\Big|_{z=\psi_{i;s}(x^i;\omega)}\overline{A}_{i;s}(x^i;t_i)}{\underline{A}_{i;s}(x^i;t_i)+\overline{A}_{i;s}(x^i;t_i)}
\end{aligned}
$$

is continuous in (x^i,t_i) for almost all $\omega\in\Omega$, (b) $F(x^i;t_i;\bullet)$ is dominated by an integrable function for fixed $(x^i,t_i)\in X^i\times\mathcal{T}_i$, and (c) the samples $\{\omega^\ell\}_{\ell=1}^L$ are independent and identically distributed. All three conditions are valid by the properties of $H_{i;\lambda_i;s}$ and $\psi_{i;s}$. Similarly, since

$$
\begin{aligned}
&\frac{\partial\varphi^L_{i;s}(x^i;t_i)}{\partial t_i}\\
&\quad=\frac{\dfrac{1}{L}\displaystyle\sum_{\ell=1}^L\left[\frac{\partial H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\omega^\ell);t_i;\underline{\gamma}_i)}{\partial t_i}\underline{A}^L_{i;s}(x^i;t_i)+\frac{\partial H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\omega^\ell);t_i;\overline{\gamma}_i)}{\partial t_i}\overline{A}^L_{i;s}(x^i;t_i)\right]}{\underline{A}^L_{i;s}(x^i;t_i)+\overline{A}^L_{i;s}(x^i;t_i)}
\end{aligned}
$$

is a sample average approximation of

$$
\begin{aligned}
&\frac{\partial\varphi_{i;s}(x^i;t_i)}{\partial t_i}\\
&\quad=\frac{\mathbb{E}\left[\dfrac{\partial H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\widetilde{\omega});t_i;\underline{\gamma}_i)}{\partial t_i}\right]\underline{A}_{i;s}(x^i;t_i)+\mathbb{E}\left[\dfrac{\partial H_{i;\lambda_i;s}(\psi_{i;s}(x^i;\widetilde{\omega});t_i;\overline{\gamma}_i)}{\partial t_i}\right]\overline{A}_{i;s}(x^i;t_i)}{\underline{A}_{i;s}(x^i;t_i)+\overline{A}_{i;s}(x^i;t_i)},
\end{aligned}
$$

we deduce

$$
\mathbb{P}\left\{\lim_{L\to\infty}\left[\sup_{(x^i,t_i)\in X^i\times\mathcal{T}_i}\left|\frac{\partial\varphi^L_{i;s}(x^i;t_i)}{\partial t_i}-\frac{\partial\varphi_{i;s}(x^i;t_i)}{\partial t_i}\right|\right]=0\right\}=1,
$$

completing the proof of the limit (48). □

An immediate consequence of (48) is that there exists a subset $\Omega_0\subset\Omega$ with $\mathbb{P}(\Omega_0)=0$ such that for all $\omega\in\widehat{\Omega}_0\triangleq\Omega\setminus\Omega_0$, for every $\varepsilon>0$, there exists a positive integer $L(\varepsilon,\omega)$ such that for all L and $L'\geq L(\varepsilon,\omega)$

$$
\sup_{(x^i,t_i)\in X^i\times\mathcal{T}_i}\left\|\nabla_{x^i}\varphi^L_{i;s}(x^i;t_i)-\nabla_{x^i}\varphi^{L'}_{i;s}(x^i,t_i)\right\|\leq\varepsilon
$$

and

$$
\sup_{(x^i,t_i)\in X^i\times\mathcal{T}_i}\left|\frac{\partial\varphi^L_{i;s}(x^i;t_i)}{\partial t_i}-\frac{\partial\varphi^{L'}_{i;s}(x^i;t_i)}{\partial t_i}\right|\leq\sqrt{s}\,\varepsilon,
$$

where $\varphi^{L}_{i;s}(x^i;t_i)$ and $\varphi^{L'}_{i;s}(x^i;t_i)$ pertain to the given realization ω of the samples in these two random functions.

Theorem 10 *Suppose that the set* $\mathbf{X}$ *is compact. Assume that each function* $H_{i,\lambda_i;s}$ *satisfies conditions (H1), (H2), and* (H^{apprx}) *and that the recourse functions* $\psi_i(x^i,\omega)$ *satisfies the assumptions of Proposition 2 for all* $\omega \in \Omega$. *Let* $\{L_\nu\}$ *be an increasing sequence of positive scalars tending to* ∞. *For each* ν, *let* $\{\omega^{\nu,1},\ldots,\omega^{\nu,L_\nu}\}$ *be* L_ν *independently and identically distributed samples of the random variable* $\widetilde{\omega}$ *and let* $\{(x^\nu,t^\nu)\}$ *be a sequence of iterates obtained by iteratively solving the SPs* (47) *corresponding to these samples. Then* $\{(x^\nu,t^\nu)\}$ *converges almost surely to the unique NE of the game* $\mathcal{G}_s^{\text{srMD}}$.

Proof In the proof below, we fix an arbitrary realization $\omega \in \widehat{\Omega}_0$ of the random variable $\widetilde{\omega}$ and its samples. Let $\varepsilon > 0$ be arbitrary. Let ν_0 be such that $L_\nu \geq L(\varepsilon,\omega)$ for all $\nu \geq \nu_0$. For each $\nu \geq \nu_0 + 1$, similar to the proof of Theorem 4, we can deduce the following two inequalities:

$$\begin{aligned}
&(x^{\nu;i} - x^{\nu+1;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) + \nabla_{x^i}\varphi^{L_\nu}_{i;s}(x^{\nu+1;i}, t_i^{\nu+1}) \right] \\
&\quad + (x^{\nu;i} - x^{\nu+1;i})^T (x^{\nu+1;i} - x^{\nu;i}) \\
&\quad + (t_i^{\nu} - t_i^{\nu+1}) \left(s\, t_i^{\nu+1} + \frac{\partial \varphi^{L_\nu}_{i;s}(x^{\nu+1;i}; t_i^{\nu+1})}{\partial t_i} \right) \geq 0
\end{aligned}$$

and

$$\begin{aligned}
&(x^{\nu+1;i} - x^{\nu;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu;i}, x^{\nu-1;-i}) + \nabla_{x^i}\varphi^{L_{\nu-1}}_{i;s}(x^{\nu;i}, t_i^{\nu}) \right] \\
&\quad + (x^{\nu+1;i} - x^{\nu;i})^T (x^{\nu;i} - x^{\nu-1;i}) \\
&\quad + (t_i^{\nu+1} - t_i^{\nu}) \left(s\, t_i^{\nu} + \frac{\partial \varphi^{L_{\nu-1}}_{i;s}(x^{\nu;i}; t_i^{\nu})}{\partial t_i} \right) \geq 0.
\end{aligned}$$

Adding these two inequalities, we obtain

$$\begin{aligned}
&(x^{\nu;i} - x^{\nu+1;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) - \nabla_{x^i}\theta_i(x^{\nu;i}, x^{\nu-1;-i}) \right] \\
&\quad + (x^{\nu;i} - x^{\nu+1;i})^T \left[\nabla_{x^i}\varphi^{L_\nu}_{i;s}(x^{\nu+1;i}; t_i^{\nu+1}) - \nabla_{x^i}\varphi^{L_{\nu-1}}_{i;s}(x^{\nu;i}; t_i^{\nu}) \right] \\
&\quad + (t_i^{\nu} - t_i^{\nu+1}) \left[s\,(t_i^{\nu+1} - t_i^{\nu}) + \left(\frac{\partial \varphi^{L_\nu}_{i;s}(x^{\nu+1;i}; t_i^{\nu+1})}{\partial t_i} - \frac{\partial \varphi^{L_{\nu-1}}_{i;s}(x^{\nu;i}; t_i^{\nu})}{\partial t_i} \right) \right] \\
&\quad + (x^{\nu;i} - x^{\nu+1;i})^T (x^{\nu+1;i} - x^{\nu;i}) + (x^{\nu+1;i} - x^{\nu;i})^T (x^{\nu;i} - x^{\nu-1;i}) \geq 0.
\end{aligned} \tag{49}$$

We have

$$\begin{aligned}
&\nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu;i};t_i^{\nu}) \\
&\quad = \nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1}) \\
&\qquad + \left[\nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu;i};t_i^{\nu})\right].
\end{aligned}$$

Similarly,

$$\begin{aligned}
&\frac{\partial\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} - \frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu;i};t_i^{\nu})}{\partial t_i} \\
&\quad = \frac{\partial\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} - \frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} \\
&\qquad + \left[\frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} - \frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu;i};t_i^{\nu})}{\partial t_i}\right].
\end{aligned}$$

By the convexity of the function $\varphi_{i;s}^{L_{\nu-1}}$, we have

$$\begin{aligned}
&\left(x^{\nu+1;i} - x^{\nu;i}\right)^T \left[\nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu;i};t_i^{\nu})\right] \\
&\quad + \left(t_i^{\nu+1} - t_i^{\nu}\right)\left[\frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} - \frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu;i};t_i^{\nu})}{\partial t_i}\right] \geq 0.
\end{aligned}$$

Consequently, (49) yields

$$\begin{aligned}
&(x^{\nu+1;i} - x^{\nu;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) - \nabla_{x^i}\theta_i(x^{\nu;i}, x^{\nu-1;-i})\right] \\
&\quad + \| x^{\nu+1;i} - x^{\nu;i} \|^2 + s\,(t_i^{\nu+1} - t_i^{\nu})^2 \\
&\leq (x^{\nu;i} - x^{\nu+1;i})^T \left[\nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1})\right] \\
&\quad + (x^{\nu+1;i} - x^{\nu;i})^T (x^{\nu;i} - x^{\nu-1;i}) \\
&\quad + (t^{\nu,i} - t_i^{\nu+1})\left[\frac{\partial\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} - \frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i}\right].
\end{aligned}$$

By the choice of the iteration counter ν, the two bounds:

$$\left\| \nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1}) \right\| \leq \varepsilon$$

and

$$\left| \frac{\partial\varphi_{i;s}^{L_\nu}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} - \frac{\partial\varphi_{i;s}^{L_{\nu-1}}(x^{\nu+1;i};t_i^{\nu+1})}{\partial t_i} \right| \leq \sqrt{s}\,\varepsilon$$

hold. Hence,

$$
\begin{aligned}
&(x^{\nu+1;i} - x^{\nu;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) - \nabla_{x^i}\theta_i(x^{\nu;i}, x^{\nu-1;-i}) \right] \\
&\quad + s\,(t_i^{\nu+1} - t_i^{\nu})^2 + \| x^{\nu+1;i} - x^{\nu;i} \|^2 \\
&\leq \varepsilon \left[\| x^{\nu+1;i} - x^{\nu;i} \| + \sqrt{s}\,|\, t_i^{\nu+1} - t_i^{\nu} \,| \right] \\
&\quad + (x^{\nu+1;i} - x^{\nu;i})^T (x^{\nu;i} - x^{\nu-1;i})
\end{aligned}
$$

Following the derivation in Theorem 4, we deduce the existence of a (contraction) scalar δ in $(0, 1)$ derived from condition (DD) on the family of first-stage objectives $\{\theta_i\}_{i=1}^N$, such that for all $\nu \geq \nu_0 + 1$,

$$
\begin{aligned}
e_{\nu+1} &\leq \delta\, e_\nu + \varepsilon, \quad \text{where } e_{\nu+1} \\
&\triangleq \left\| \begin{pmatrix} \| x^{\nu+1;1} - x^{\nu;1} \| + \dfrac{\sqrt{s}}{1+\gamma_{11}}\,|\, t_1^{\nu+1} - t_1^{\nu} \,| \\ \vdots \\ \| x^{\nu+1;N} - x^{\nu;N} \| + \dfrac{\sqrt{s}}{1+\gamma_{NN}}\,|\, t_N^{\nu+1} - t_N^{\nu} \,| \end{pmatrix} \right\| .
\end{aligned}
$$

Iterating on this recursion, we deduce

$$
e_{\nu+1} \;\leq\; \delta^{\,\nu-\nu_0+1}\, e_{\nu_0} + \varepsilon \sum_{k=0}^{\nu_0-1} \delta^k \;\leq\; \delta^{\nu-\widehat{\nu}+1}\, e_{\nu_0} + \frac{\varepsilon}{1-\delta}\,.
$$

Hence $\limsup_{\nu\to\infty} e_\nu \;\leq\; \dfrac{\varepsilon}{1-\delta}$. Since ε is arbitrary, it follows that $\lim_{\nu\to\infty} e_\nu = 0$. To complete the proof, let (x^∞, t^∞) be an accumulation point of the sequence $\{(x^\nu, t^\nu)\}$. The conclusion that (x^∞, t^∞) is a Nash equilibrium of the game $\mathcal{G}_s^{\text{srMD}}$ follows from three observations: (a) the variational inequality below holds for all $i = 1, \ldots, N$ and all ν:

$$
\begin{aligned}
&(x^i - x^{\nu+1;i})^T \left[\nabla_{x^i}\theta_i(x^{\nu+1;i}, x^{\nu;-i}) + \nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i}, t_i^{\nu+1}) \right] \\
&\quad + (x^i - x^{\nu+1;i})^T (x^{\nu+1;i} - x^{\nu;i}) \\
&\quad (t_i - t_i^{\nu+1}) \left(s\, t_i^{\nu+1} + \frac{\partial \varphi_{i;s}^{L_\nu}(x^{\nu+1;i}; t_i^{\nu+1})}{\partial t_i} \right) \geq 0, \ \forall\, (x^i, t_i) \in X^i \times \mathcal{T}_i,
\end{aligned}
$$

(b) the identity

$$
\begin{aligned}
&\nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i}, t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}(x^{\infty;i}, t_i^{\infty}) \\
&\quad = \left[\nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i}, t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}(x^{\nu+1;i}, t_i^{\nu+1}) \right] \\
&\qquad + \left[\nabla_{x^i}\varphi_{i;s}(x^{\nu+1;i}, t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}(x^{\infty;i}, t_i^{\infty}) \right],
\end{aligned}
$$

shows that $\nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i}, t_i^{\nu+1})$ converges to $\nabla_{x^i}\varphi_{i;s}(x^{\infty;i}, t_i^{\infty})$ because the differences within the two square brackets on the right-hand side converge to zero, and (c) a limit similar to that in (b) holds for $\dfrac{\partial\varphi_{i;s}^{L_\nu}(x^{\nu+1;i}; t_i^{\nu+1})}{\partial t_i}$. Finally, the fact that the accumulation point (x^∞, t^∞) is actually the limit of the entire sequence $\{(x^\nu, t^\nu)\}$ can be proved in two steps: (a) take another accumulation point, say $(\widehat{x}^\infty, \widehat{t}^\infty)$, which must necessarily be a NE of the game $\mathcal{G}_s^{\mathrm{srMD}}$ also, and (b) repeat the above proof with (x^∞, t^∞) and $(\widehat{x}^\infty, \widehat{t}^\infty)$ taking the place of $(x^{\nu+1}; t^{\nu+1})$ and $(x^\nu; t^\nu)$, respectively. These two steps will then yield the equality of these two accumulation points.

In summary, we have therefore shown that for all realizations $\omega \in \widehat{\Omega}_0$, the sequence $\{(x^\nu, t^\nu)\}$ converges to the unique NE of the game $\mathcal{G}_s^{\mathrm{srMD}}$; hence this random sequence converges almost surely to the said equilibrium. □

Remarks A noteworthy remark is that the t-variable has been regularized in the game $\mathcal{G}_s^{\mathrm{srMD}}$; thus this variable is also unique in a NE of this game. This is different from the previous remark made after Theorem 4 about the un-regularized game $\mathcal{G}^{\mathrm{prMD}}$ where the t-variable has not been regularized; thus no uniqueness of this variable in a NE was asserted. □

To close this discussion, we remark that the above proof reveals the complication in dealing with coupled recourse functions; indeed, in this case of coupled second-stage recourse, we would need to evaluate a difference such as

$$\nabla_{x^i}\varphi_{i;s}^{L_\nu}(x^{\nu+1;i}; x^{\nu;-i}; t_i^{\nu+1}) - \nabla_{x^i}\varphi_{i;s}^{L_{\nu-1}}(x^{\nu;i}; x^{\nu-1;-i}; t_i^{\nu})$$

in which x^{-i} appears in both terms. While this may be possible, the details are best left for another study that will aim to investigate the convergence of the best-response scheme under relaxed differentiability requirements.

Concluding remarks Starting from a review of existing approaches for dealing with the Nash equilibrium problem under uncertainty, we have introduced the two-stage SNEP with risk-averse players and quadratic recourse function and examined its iterative solution by best-response type algorithms. A deep understanding of the functional properties of the recourse functions is a starting point of the analysis. Setting aside the issue of sampling, the case of private recourse can be treated relatively easily by solving stochastic programming subproblems. When sampling and best response are jointly applied, the convergence of the overall scheme is in jeopardy. In order to establish the convergence of the solution algorithm resulting from the joint application of sampling and best response, two preparatory treatments are introduced: smoothing and regularization. Even with these and the background preparations, the convergence proof of a sampled best-response scheme remains elusive when coupling of players' variables occurs in their recourse functions. This is the limitation of the analysis in this paper; it is not enough to cover the complete two-stage SNEP where the first-stage and second-stage constraints have the players' decision variables coupled. Further study is needed to address this extended case as well as other possible approaches that may include a Lagrangian relaxation of these coupled constraints. The question of how the progressive hedging approach derived from the basic non-anticipativity property in SP

can be applied to solve these problems remains to be studied. Lastly, implementation of the proposed solution methods and their application to realizations of these games are important topics to be investigated in the future. Overall, our study has barely scratched the surface of this vast topic that combines competitive decision making, uncertainty, and second-stage recourse all in a game-theoretic context.

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